DR64 Primary HSI C

Prepared by:	Chaithra shetty
Approved by:	Haim Giladi / Moshe Shalev
Version	0.1
Release date	11-06-2026

1 Introduction

This document describes a technical detail of the Primary Board B2. This document describes both production intent hardware and Debug hardware, for the production vs Debug changes please refer Chapter 9.

2 C Board Connectivity

HW Architecture block diagram

Pri_SV62 + TDA4+A-PHY Master System Arch Diagram B2 V1.0 250624.pdf

3 MCU Block Diagram

SM-68470 - Detailed view of MCU connectivity

Pri_SV62 + TDA4+A-PHY Master System Arch Diagram B2 V1.0 250624.pdf [In Review]

4 S32G3

4.1 S32G3 Power tree diagram

Attached the power tree of Primary B2. (needs to be updated to C)

Power_Tree_And_Thermal_Map_SV62_CH63_DR64_C_Sample

4.2 S32G3 has two main power supplies:

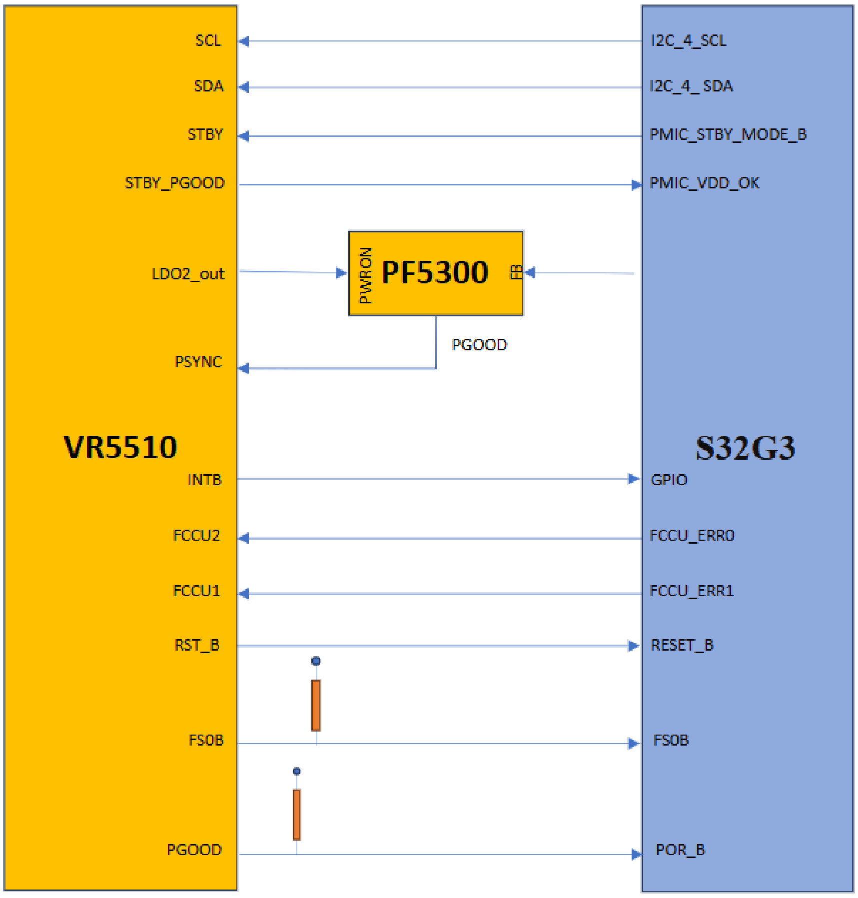
Power supplies

- PMIC VR5510 with several outputs, used for I/Os & peripherals.
- Buck regulator PF5300 used for 0.8V MCU core voltage

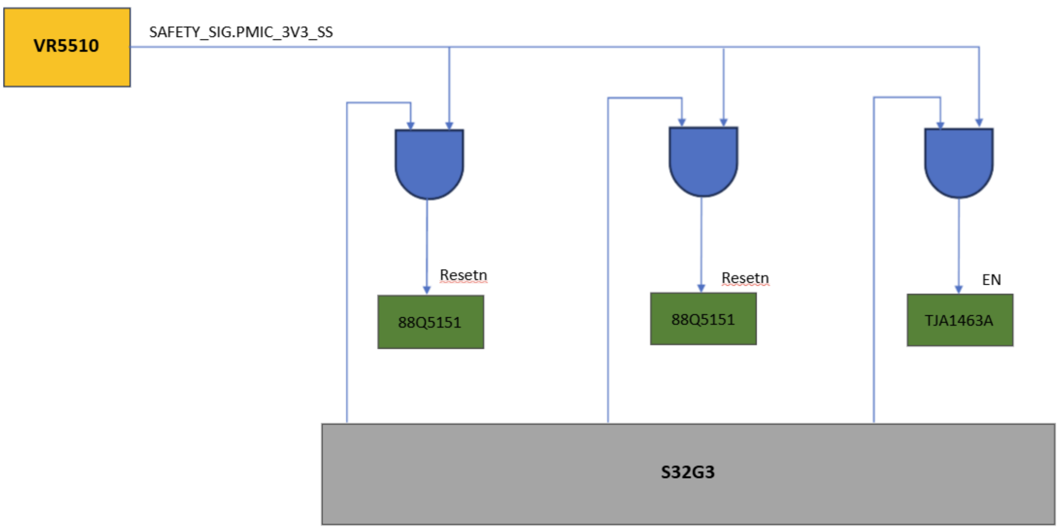
4.3 S32G3 & Peripherals Power Rails

Device	QTY	MCU_3V3	MCU_0V8_STBY	MCU_1V8	MCU_1V1	MCU_0V8_VDD_CORE	MCU_1V8_ANA	5V_CAN
MCU	x 1	3.3V	0.8V	1.8V	1.1V	0.8V	1.8V	
LPDDR4	x 1			1.8V	1.1V			
NOR Flash	x 1			1.8V				
eMMC	x 1	3.3V		1.8V				
CAN Trans.	x 7	3.3V						5V
Misc.	x 10	3.3V						

4.4 S32G3 PMIC Connectivity - HW Safety



4.5 PMIC FS0B Connectivity

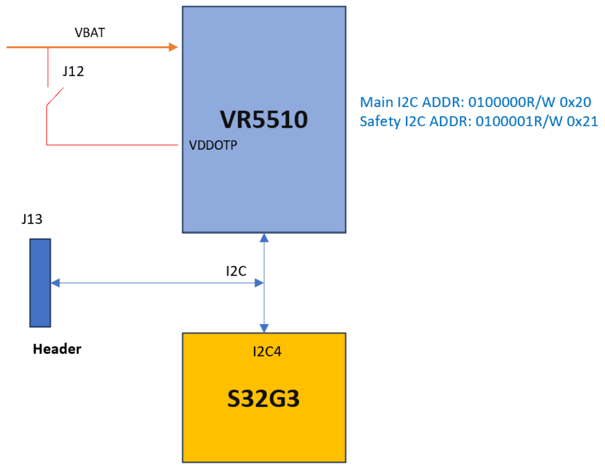


4.6 PMIC

4.6.1 Configuration

MCU PMIC is a Pre-programmed device, P.N: MVR5510AMDALES. Below is the pre-configured data
R-MVR5510AMDALES.pdf

4.6.2 Debug Mode



The VR5510 enters in Debug mode with the following sequence:

- Apply VDDOTP pin = VDDOTP.
- Apply VSUP1/2 > V_{SUP_UVH} and PWRON1 > PWRON1$_{VIH}$ or PWRON2 > PWRON2$_{VIH}$.
- The device now starts in Debug mode, ready for debugging or OTP programming.
- Apply VDDOTP = 0 V to turn on the device with the modified configuration.

4.7 S32G Hardware Configuration (BOOTMOD) pins

FUSE_ SEL	BMOD E1	BMOD E2	Boot mode	Available Boot interface	Use case
0	0	0	Serial boot + XOSC in Differential mode	Serial Communication interfaces: UART and CAN	Production configuration for a unprogrammed chip
0	0	1	Serial boot + XOSC in Crystal mode or Bypass mode	Serial Communication interfaces: UART and CAN	Production configuration for a unprogrammed chip
0	1	0	Boot from RCON	Memory Interfaces: QSPI, SD card, MMC, eMMC	Development configuration
1	0	X	Boot from fuses	Memory Interfaces: QSPI, SD card, MMC, eMMC	Production configuration for a programmed chip
1	1	0	Serial boot	Serial Communication interfaces: UART and CAN	Debug of a programmed chip
X	1	1	Reserved		Reserved

4.7.1 BOOTMOD Pins – SW1 Debug

Peripheral	Signal	Pin/Port#	Net Name
BOOT	mode, 0	[W10] PA_02	BOOTMOD0
BOOT	mode, 1	[W11] PA_03	BOOTMOD1

NOTE: If the switch is closed the value of the BOOTMODE will be 1.

4.7.2 MCU Development Config.

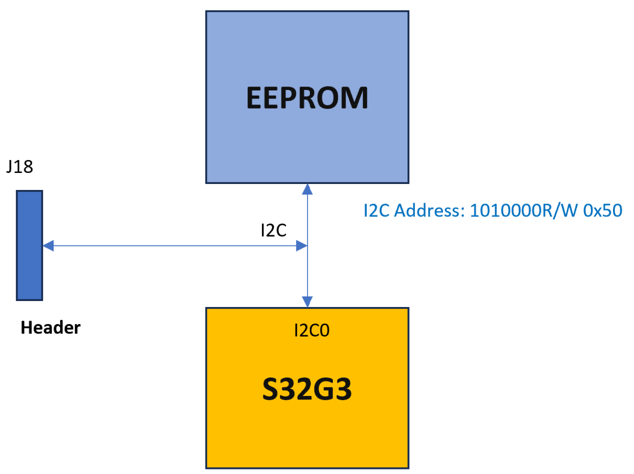
4.7.2.1 BOOT STRAP

BOOT_CFG2 [4]	PA_02	PA_03	Mode		RCON8/PB_01	
FUSE_SEL	BOOTMOD1	BOOTMOD2			eePROM/Serial	Pins/Parallel
0	1	0	Boot from RCON	→	1	0

4.8 Serial BOOT

Peripheral	Signal	Pin/Port#	Net Name
I2C_0	sda, 0	[W12] PB_00	MCU_IF.3v3_I2C0_SDA
I2C_0	scl, 0	[E7] PB_01	MCU_IF.3v3_I2C0_SCL

4.8.1 I2C eePROM



4.8.2 Description

- eePROM P.N: AT24C01C
- I2C tool like Aardvark shall be used to download the eePROM.
- J18 is connected to I2C0

SM-80469 - The first 4 bytes must be set:

- NOR Flash - 0x0000
- eMMC – 0x0060
- This eePROM can be downloaded by JTAG too.
- Board type details can be accessed from the eePROM
- Board Revision details can be accessed from the eePROM

[Approved **]**

4.9 MI information

Reference: MCU MI Software Requirements SV62, MI information

Check for DR64 C sample

SM-80470 - MCU MI data shall be written to NOR flash with an allocated size of at least 1024 Bytes. NOR Flash 0x0FFE0000 - 0x0FFFFFFF (128 KB) address range is allocated for MI block **[** Approved **]**

SM-80471 - Board Name
The board name is a text describing the board. String is zero terminated.

- Parameter name: board_name
- Type: str
- size in bytes: 64
- Min value: 1 char
- Max value: 63 chars

[Approved **]**

SM-80472 - Board_ID
This board_id value should be the dec value of the VW_ECU_HW_version 3 first bytes value.

- Parameter name: board_id
- Type: u32, dec value
- size in bytes: 4
- Min value: 0
- Max value: 32 bits

[Approved **]**

SM-80473 - Board revision
The MCU MI shall contain the ECU board revision. which describes the board **revision** for the board type indicated in the board_id field (e.g. B sample rev1, B sample rev2).

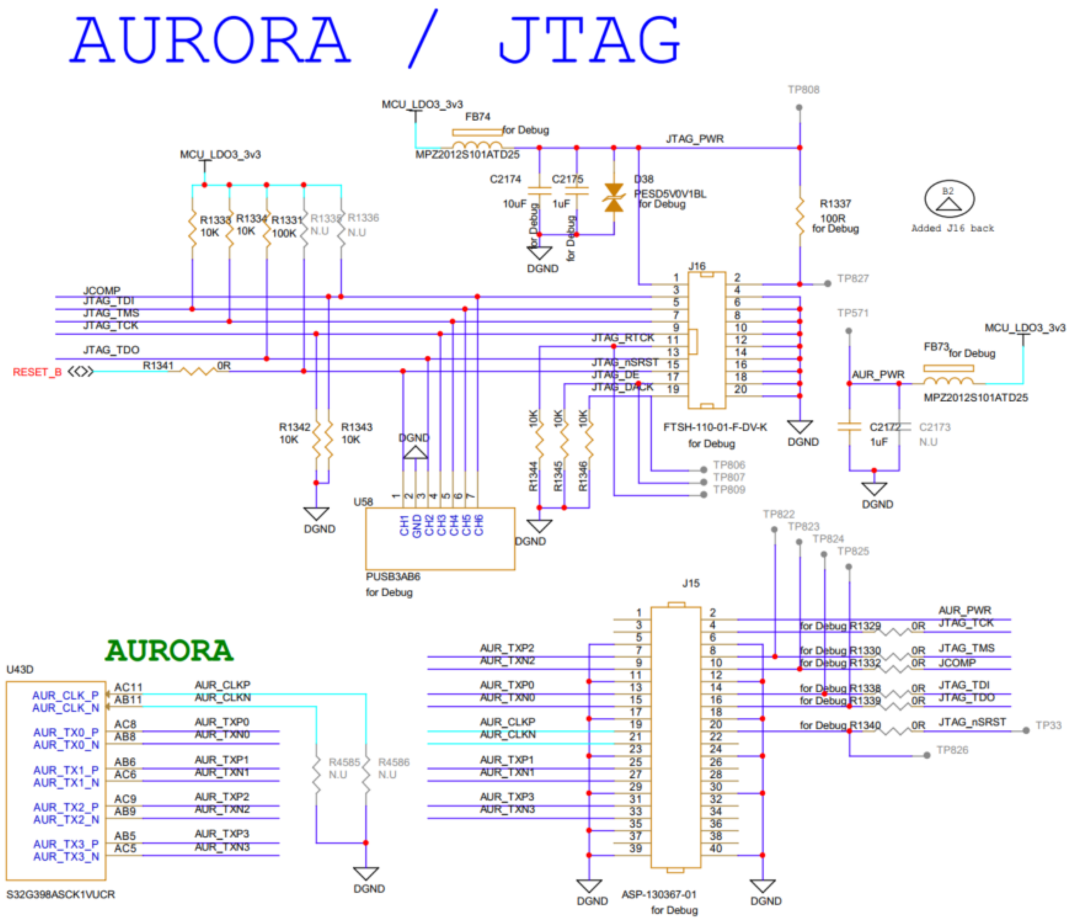
- Parameter name: board_rev
- Type: u32, dec value
- size in bytes: 4
- Min value: 0
- Max value: 32 bits

[Approved **]**

4.10 Debug S32G

4.10.1 Lauterbach/S32DBGPROBE - JTAG programmer device

For C board, both J15 and J16 exists and can support JTAG like following:



J16:

Connector Pin	Signal	Pin/Port#	Signal Name
5	JTAG test data input	[V7] PA_00	JTAG_TDI
13	JTAG test data output	[AA7] PA_01	JTAG_TDO
9	JTAG test clock	[W9] PA_04	JTAG_TCK
7	JTAG test mode selection	[U7] PA_05	JTAG_TMS
3	jcomp Pin	[W7] JCOMP	JCOMP
15	JTAG reset pin	RESET_B	JTAG_nSRST

J15:

Connector Pin	Signal	Pin/Port#	Signal Name
14	JTAG test data input	[V7] PA_00	JTAG_TDI
16	JTAG test data output	[AA7] PA_01	JTAG_TDO

Connector Pin	Signal	Pin/Port#	Signal Name
4	JTAG test clock	[W9] PA_04	JTAG_TCK
8	JTAG test mode selection	[U7] PA_05	JTAG_TMS
10	jcomp Pin	[W7] JCOMP	JCOMP
20	JTAG reset pin	RESET_B	JTAG_nSRST
7	Aurora Transmit lane 2	[AC9] AUR_TXP2	AUR_TXP2
9	Aurora Transmit lane 2	[AB9] AUR_TXN2	AUR_TXN2
13	Aurora Transmit lane 0	[AC8] AUR_TXP0	AUR_TXP0
15	Aurora Transmit Lane 0	[AB8] AUR_TXN0	AUR_TXN0
19	Aurora reference Clock input	[AC11] AUR_CLKP	AUR_CLKP
21	Aurora reference Clock input	[AB11] AUR_CLKN	AUR_CLKN
25	Aurora Transmit Lane 1	[AC6] AUR_TXP1	AUR_TXP1
27	Aurora Transmit Lane 1	[AB6] AUR_TXN1	AUR_TXN1
31	Aurora Transmit Lane 3	[AB5] AUR_TXP3	AUR_TXP3
33	Aurora Transmit Lane 3	[AC5] AUR_TXN3	AUR_TXN3

4.10.2 Debug UART

#	Peripheral	Signal	Routed pin/signal	Label
U11	LINFlex_0	txd	[U11] PC_09	MCU_IF.3v3_MCU_CLI_UART_TX
Y12	LINFlex_0	rxd	[Y12] PC_10	MCU_IF.3v3_MCU_CLI_UART_RX
A9	LINFlex_2	txd	[A9] PK_11	DEBUG.3v3_MCU_UART_TX
C14	LINFlex_2	rxd	[C14] PK_12	DEBUG.3v3_MCU_UART_RX

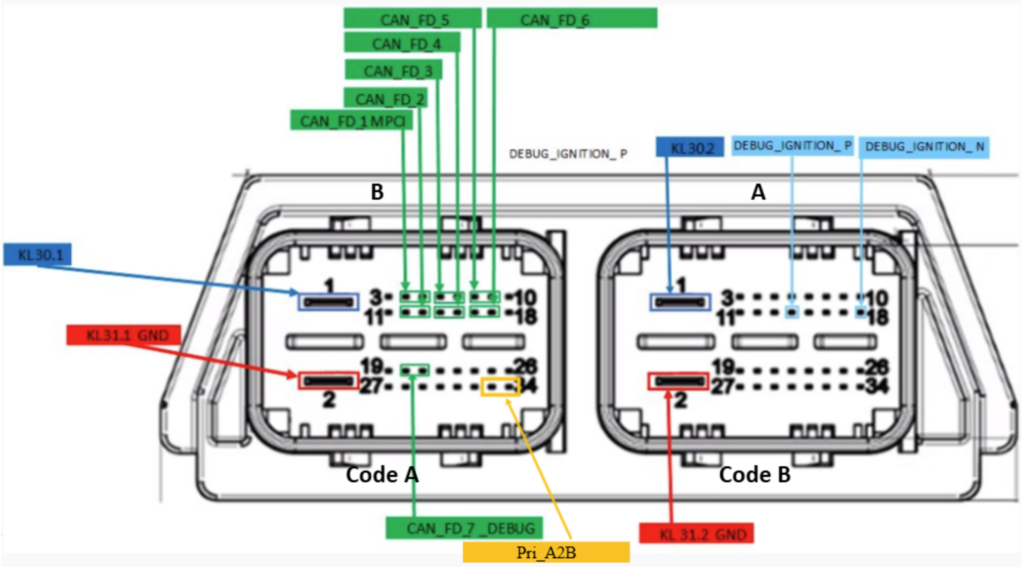
4.10.3 Debug CAN

#	Peripheral	Signal	Routed pin/signal	Label
E12	LLCE_CAN_6	txd	[E12] PJ_11	CAN.3v3_DEBUG_TX
E16	LLCE_CAN_6	rxd	[E16] PJ_12	CAN.3v3_DEBUG_RX

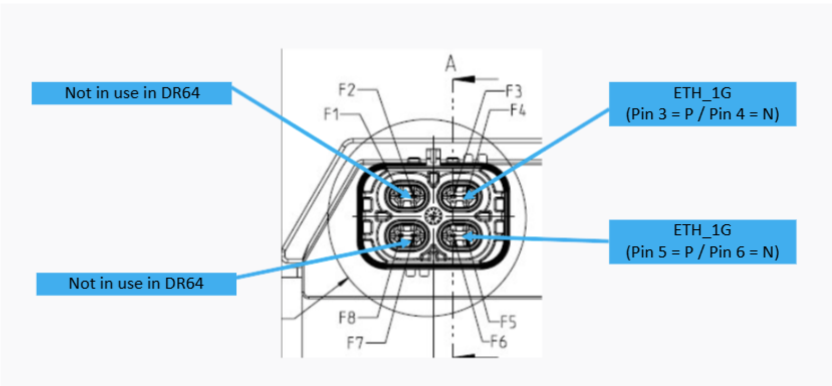
4.10.4 Debug GPIO

Test point of PB_11[G8] pin of S32G3 is used for the 1 PPS signal generation purpose, so that a high-impedance measuring instrument can be connected to it permanently.

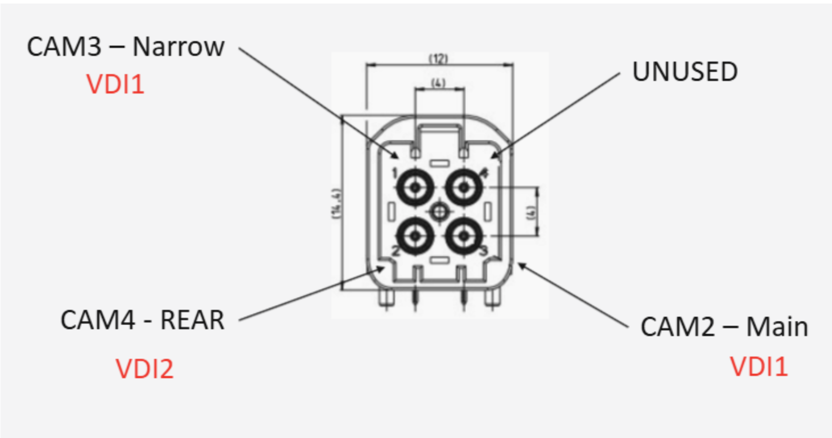
4.11 Front Panel – DR64 Main Connector



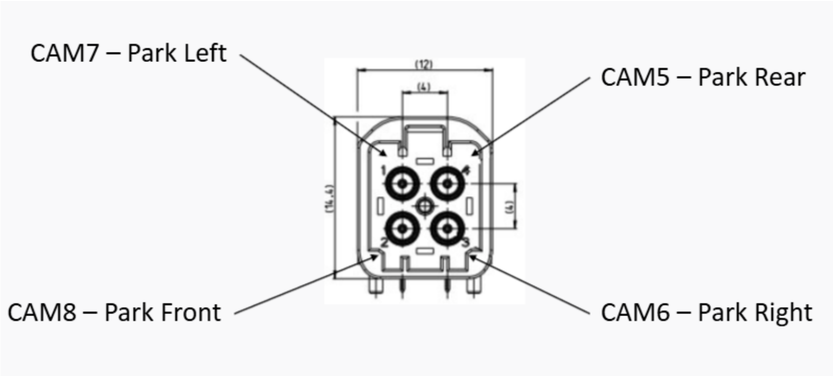
F Connector



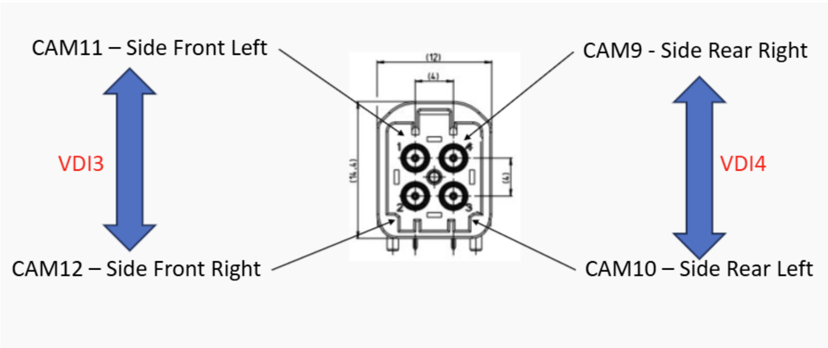
C Connector



D Connector



E Connector



4.12 U104 Configuration

Control Register

Following the successful acknowledgment of the address byte, the bus controller sends a command byte shown in

0	0	0	0	0	B2	B1	B0

CONTROL REGISTER BITS

B2	B1	B0	Command Byte (Hex)	Register	Protocol	Power-Up Default
CONTROL REGISTER BITS						
0	0	0	0x00	Input Port 0	Read byte	xxxx xxxx
0	0	1	0x01	Input Port 1	Read byte	xxxx xxxx
0	1	0	0x02	Output Port 0	Read-write byte	1111 1111
0	1	1	0x03	Output Port 1	Read-write byte	1111 1111
1	0	0	0x04	Polarity Inversion Port 0	Read-write byte	0000 0000
1	0	1	0x05	Polarity Inversion Port 1	Read-write byte	0000 0000
1	1	0	0x06	Configuration Port 0	Read-write byte	1111 1111
1	1	1	0x07	Configuration Port 1	Read-write byte	1111 1111

Configuration registers

The Configuration registers (registers 6 and 7) shown below configure the directions of the I/O pins. If a bit in this register is set to 1, the corresponding port pin is enabled as an input with a high-impedance output driver. If a bit in this register is cleared to 0, the corresponding port pin is enabled as an output.

Registers 6 And 7 (Configuration Registers)

C0	C0.7	C0.6	C0.5	C0.4	C0.3	C0.2	C0.1	C0.0
Default	1	1	1	1	1	1	1	1
C1	C1.7	C1.6	C1.5	C1.4	C1.3	C1.2	C1.1	C1.0
Default	1	1	1	1	1	1	1	1

Output Port registers

The Output Port registers (registers 2 and 3) shown below show the outgoing logic levels of the pins defined as outputs by the Configuration register. Bit values in this register have no effect on pins defined as inputs. In turn, reads from this register reflect the value that is in the flip-flop controlling the output selection, not the actual pin value.

Registers 2 And 3 (Output Port Registers)

O0	O0.7	O0.6	O0.5	O0.4	O0.3	O0.2	O0.1	O0.0
Default	1	1	1	1	1	1	1	1
O1	O1.7	O1.6	O1.5	O1.4	O1.3	O1.2	O1.1	O1.0
Default	1	1	1	1	1	1	1	1

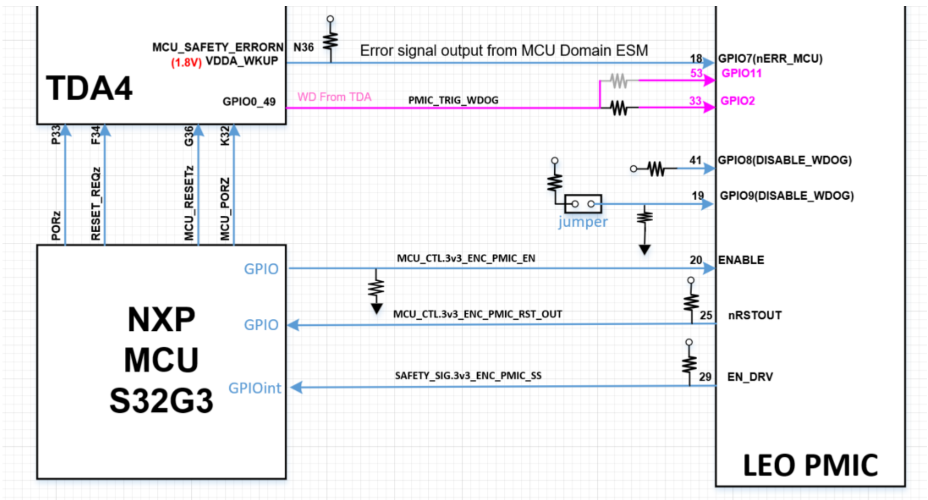
Input Port registers

The Input Port registers (registers 0 and 1) shown below reflect the incoming logic levels of the pins, regardless of whether the pin is defined as an input or an output by the Configuration register. It only acts on read operation. Writes to these registers have no effect. The default value, X, is determined by the externally applied logic level

Registers 0 And 1 (Input Port Registers)

I0	I0.7	I0.6	I0.5	I0.4	I0.3	I0.2	I0.1	I0.0
Default	X	X	X	X	X	X	X	X
I1	I1.7	I1.6	I1.5	I1.4	I1.3	I1.2	I1.1	I1.0
Default	X	X	X	X	X	X	X	X

4.13 TDA4 RESET Diagram



TDA4 Reset IF					
Device/Pin		Dir. S32G3	Net Name	Description	Notes
U43.A8	INT GPIO	I	MCU_CTL.3v3_ENC_PMIC_RST_OUT	LEO PMIC	PMIC Output
U103.10	I2C#0 Add. 0x76	O	ENC.3v3_RESET_REQZ	Main Domain external Warm Reset request	TDA4 Input
U43.P23	GPIO	O	ENC.1v8_PORZ	SoC PORz Reset Signal	TDA4 Input
U43.L21	GPIO	O	ENC.1v8_MCU_PORZ	MCU Domain Cold Reset	TDA4 Input
U103.11	I2C#0 Add. 0x76	O	ENC.3v3_MCU_RESETZ	MCU Domain Warm Reset	TDA4 Input

TDA4 Modes

TDA4 UART_MODE:

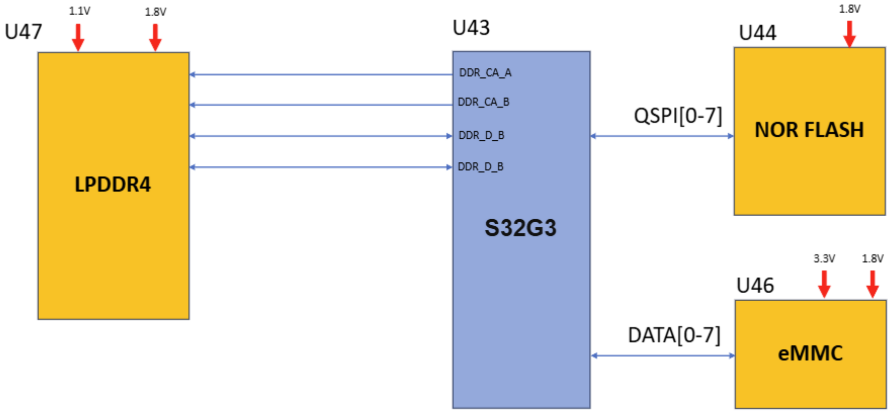
- ENC.3v3_MCU_BOOTMODE3- High
- ENC.3v3_MCU_BOOTMODE4- High
- ENC.3v3_MCU_BOOTMODE5- High
- ENC.1v8_SoC_BOOTMODE0- Low
- ENC.3v3_SoC_BOOTMODE4- Low
- ENC.3v3_SoC_BOOTMODE5- Low
- ENC.3v3_SoC_BOOTMODE6- Low

TDA4 NOR_MODE:

- ENC.3v3_MCU_BOOTMODE3- High
- ENC.3v3_MCU_BOOTMODE4- High
- ENC.3v3_SoC_BOOTMODE4- High
- ENC.3v3_SoC_BOOTMODE6- High
- ENC.3v3_MCU_BOOTMODE5- Low
- ENC.1v8_SoC_BOOTMODE0- Low
- ENC.3v3_SoC_BOOTMODE5- Low

4.14 MCU Memories

4.14.1 Block Diagram



4.14.2 MCU & Peripherals memories

SM-71385 -

Device	P.N	Manuf.	Ref. Des.	Description	notes
MCU	S32G398ASCK1VUCR	NXP	U43	15MB RAM, 8 Cores	
LPDDR4	MT53E1G32D2FW-046 AAT:C	Micron	U47	4GB	
NOR Flash	MX66UM2G45GXRR00	Macronix	U44	256MB	
eMMC	MTFC128GAZAQJP-AAT	Micron	U46	128GB	

[✅ Approved **]**

4.14.3 Information of Memories

SM-71386 -

			Device ID [h]				Notes
Device	P.N	MNF ID [h]	Byte2	Byte3	Sector Size	no. of Sectors	
NOR Flash	MX66UM2G45GXRR00	C2	80	3C	4KB	64K	
eMMC	MTFC128GAZAQJP-AAT	15	-		512B	250M	

[✅ Approved **]**

SM-80480 - NOR memory frequency shall be set to 100MHz due to stability issues at low temperature. **[**✅ Approved **]**

SM-80481 -

Device	P.N	MNF ID [h]	Rev ID [h]	Page Size [bytes]
LPDDR4	MT53E1G32D2FW-046 AAT:C	FF	3	2048

[✅ Approved **]**

SM-80482 - pSLC mode should be used in eMMC for the following reasons

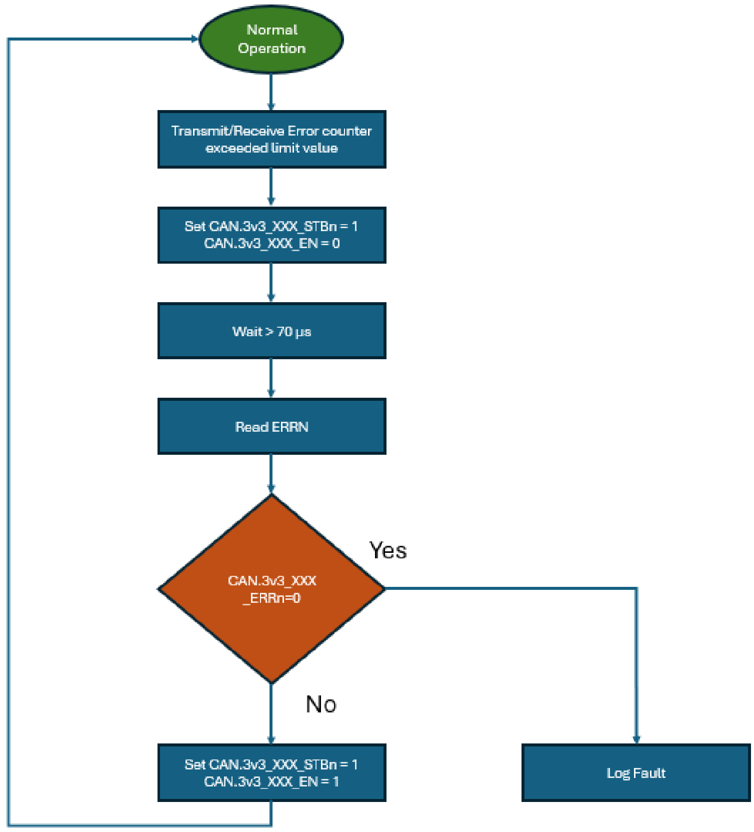
- avoid data corruption, even with power loss.
- for better data retention.

The DAF partition and partitions identified with potential risk of crossing the supported write/erase cycles (when used in MLC mode) during vehicle life time needs to be in pSLC mode. **[**✅ Approved **]**

4.15 MCU CANs

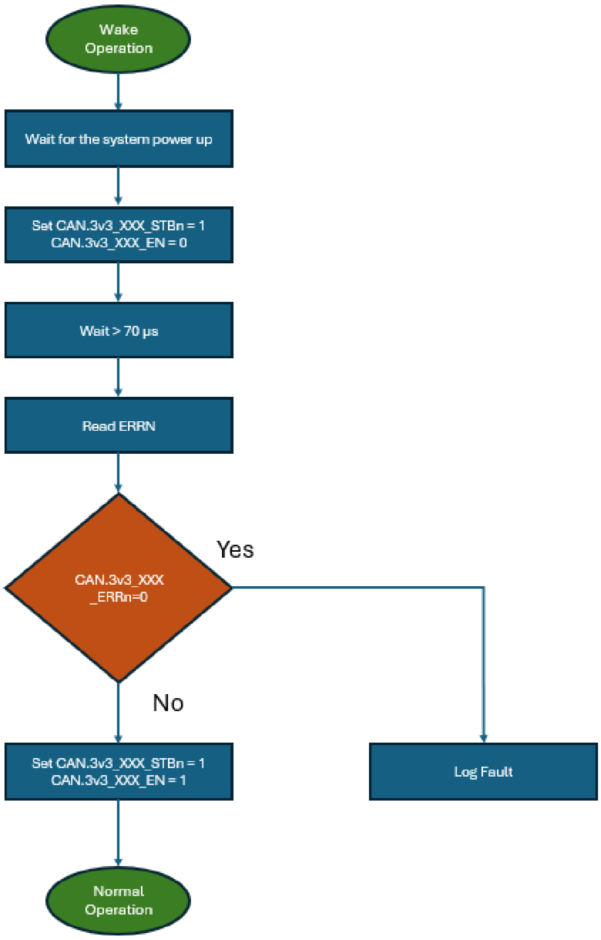
For HW Diagnostics of CAN please refer Chapter 4.1 *SuppliersManagement/HW_SW Interface/HWDIAG*

SM-80484 - Flow chart for the CAN error checking during the Normal operation



[✔ Approved]

SM-80485 - Flow chart for the CAN error checking after power up. (Self test)



[✓ Approved]

4.15.1 Flex_CAN

Peripheral	Routed pin/signal	Label	Connector Name	VOBES Name	Description
CAN_1	[AA10] PB_09	CAN.3v3_B2B_TX			B2B
CAN_1	[E6] PB_14	CAN.3v3_B2B_RX			
CAN_0	[F15] PC_11	CAN.3v3_HCP5_RX	B4: CAN.HCP5_H B5: CAN.HCP5_L	CAN_FD_1_MCPI	HCP5
CAN_0	[G11] PC_12	CAN.3v3_HCP5_TX			

4.15.2 LLCE_CAN

4.15.2.1 RADARs

Peripheral	Routed pin/signal	Label	Connector Name	VOBES Name
LLCE_CAN_1	[B11] PJ_01	CAN.3v3_RADAR2_TX	B6: CAN.RADAR2_H B7: CAN.RADAR2_L	NA for DR64
LLCE_CAN_1	[D17] PJ_02	CAN.3v3_RADAR2_RX		

Peripheral	Routed pin/signal	Label	Connector Name	VOBES Name
LLCE_CAN_2	[D10] PJ_03	CAN.3v3_RADAR3_TX	B16: CAN.RADAR3_H B17: CAN.RADAR3_L	NA for DR64
LLCE_CAN_2	[C15] PJ_04	CAN.3v3_RADAR3_RX		
LLCE_CAN_3	[C11] PJ_05	CAN.3v3_RADAR4_TX	B12: CAN.RADAR4_H B13: CAN.RADAR4_L	NA for DR64
LLCE_CAN_3	[D15] PJ_06	CAN.3v3_RADAR4_RX		
LLCE_CAN_7	[A11] PJ_13	CAN.3v3_RADAR5_TX	B14: CAN.RADAR5_H B15: CAN.RADAR5_L	NA for DR64
LLCE_CAN_7	[C16] PJ_14	CAN.3v3_RADAR5_RX		
LLCE_CAN_15	[C9] PK_13	CAN.3v3_RADAR1_TX	B8: CAN.RADAR1_H B9: CAN.RADAR1_L	NA for DR64
LLCE_CAN_15	[B13] PK_14	CAN.3v3_RADAR1_RX		

4.15.2.2 EyeQs

Pin#	Peripheral	Signal	Pin/Port#	Net Name	
E10	LLCE_CAN_13	txd	[E10] PK_09	EQ_IOs.3v3_MCU_EQ0_TX	EyeQ 0
E14	LLCE_CAN_13	rxd	[E14] PK_10	EQ_IOs.3v3_MCU_EQ0_RX	
F11	LLCE_CAN_9	txd	[F11] PK_01	EQ_IOs.3v3_MCU_EQ1_TX	EyeQ1
D14	LLCE_CAN_9	rxd	[D14] PK_02	EQ_IOs.3v3_MCU_EQ1_RX	

4.16 MCU UARTs

#	Peripheral	Signal	Routed pin/signal	Label	
G10	LLCE_LIN_2	txd	[G10] PC_05	ENC.3v3_UART_RX_R	TDA4
A14	LLCE_LIN_2	rxd	[A14] PC_06	ENC.3v3_UART_TX_R	
F10	LLCE_LIN_0	txd	[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	EyeQ_0 (A)
A12	LLCE_LIN_0	rxd	[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	EyeQ_0 (A)
B9	LLCE_LIN_3	txd	[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	EyeQ_1 (B)
G13	LLCE_LIN_3	rxd	[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	EyeQ_1 (B)

4.17 MCU ADC Monitor

4.17.1 Configuration

SM-80490 - Configuration for the S32G can be defined by below method

- Base address of ADC are:
  - ADC_0 base address: 401F_8000h
  - ADC_1 base address: 402E_8000h
- MCR configuration register with Offset: 0h is used for the main configuration
- Analog Watchdog Threshold (THRHLR**n**) register is used to define the upper thresholds [16:27] and lower thresholds [0:11]
  - THRHLR0 Offset: 60h
  - THRHLR1 Offset: 64h
  - THRHLR2 Offset: 68h

- THRHLR3 Offset: 6Ch
- THRHLR4 Offset: 280h
- THRHLR5 Offset: 284h
- THRHLR6 Offset: 288h
- THRHLR7 Offset: 28Ch
- Channel Interrupt Mask 0 (CIMR0) with Offset: 24h and Channel Interrupt Mask 1 (CIMR1) with Offset: 28h is used to enable and disable the register
- Watchdog Threshold Interrupt Status (WTISR) with Offset: 30h corresponds to the interrupt generated on the converted value being higher than the programmed higher threshold or lower than the programmed lower threshold.

[✔ Approved]

4.17.2 ADC Monitor Voltages

SM-71398 -

#	Signal	Routed pin/signal	Label	voltage [V]			Notes	HWDIAG
				Min.	Typ.	Max.		
F23	adc0_ch, 0	[F23] ADC_CH_00	N.C	-	-	-	NA	-
E22	adc0_ch, 1	[E22] ADC_CH_01	SAFETY_SIG.RED_GND_IMON	0 0X000	0.4 0X38E	1.4 0XC71		☐ PLATFORM-HW-DEV-12716
E23	adc0_ch, 2	[E23] ADC_CH_02	VBAT_INPUT_STAGE.MAIN_HSS_VMON	0.244 0X22B	0.8 0X71C	1.8 0XFFF		☐ PLATFORM-HW-DEV-9363
D22	adc0_ch, 3	[D22] ADC_CH_03	VBAT_INPUT_STAGE.MAIN_HSS_IMON	0 0X000	0.4 0X38E	1.4 0XC71		☐ PLATFORM-HW-DEV-9366
B21	adc0_ch, 4	[B21] ADC_CH_04	VR5510_AMUX	32 Analog Channels				☐ PLATFORM-HW-DEV-9373
B22	adc0_ch, 5	[B22] ADC_CH_05	SAFETY_SIG.MAIN_GND_IMON	0 0X000	0.4 0X38E	1.4 0XC71		☐ PLATFORM-HW-DEV-12715
D23	adc0_ch, 6	[D23] ADC_CH_06	N.C	-	-	-	NA	-
C22	adc0_ch, 7	[C22] ADC_CH_07	VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON	0.244 0X22B	0.61 0X56C	1.4 0XC71		☐ PLATFORM-HW-DEV-9364
A19	adc1_ch, 0	[A19] ADC_CH_08	HEATER.HSS_MON	-	-	-	-	
B20	adc1_ch, 1	[B20] ADC_CH_09	HEATER.LSS_MON	-	-	-	-	

#	Signal	Routed pin/signal	Label	voltage [V]			Notes	HWDIAG
A20	adc1_ch, 2	[A20] ADC_CH_10	VBAT_INPUT_STAGE.RED_IN_HSS_VMON	0.244 0X22B	0.61 0X56C	1.4 0XC71		☐ PLATFORM-HW-DEV-9365
C21	adc1_ch, 3	[C21] ADC_CH_11	VBAT_INPUT_STAGE.MCU_HSS_IMON	0 0X000	0.3 0X2AA	1 0X8E3		☐ PLATFORM-HW-DEV-9367

Equation for Analog to digital conversion is: (read value)*4096/1.8 to hex [✔ Approved]

4.17.3 ADC Monitor Calculations

SM-71403 - For DR64

Net Name	Real value
VBAT_INPUT_STAGE.MAIN_HSS_VMON	(read value)*15
VBAT_INPUT_STAGE.MAIN_HSS_IMON	(read value)*40
HEATER.HSS_CURRENT	(read value)*2.01
HEATER.HSS_MON	(read value)*10.1
HEATER.LSS_MON	(read value)*10.1
VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON	[(read value)*(19.1)] +0.43
VBAT_INPUT_STAGE.RED_IN_HSS_VMON	[(read value)*(19.1)] +0.43
VBAT_INPUT_STAGE.MCU_HSS_IMON	(read value)*20
USS.LSS_MON	(read value)*22.234
SAFETY_SIG.MAIN_GND_IMON	(read value-0.25)/0.025
SAFETY_SIG.RED_GND_IMON	(read value-0.25)/0.025

[✔ Approved]

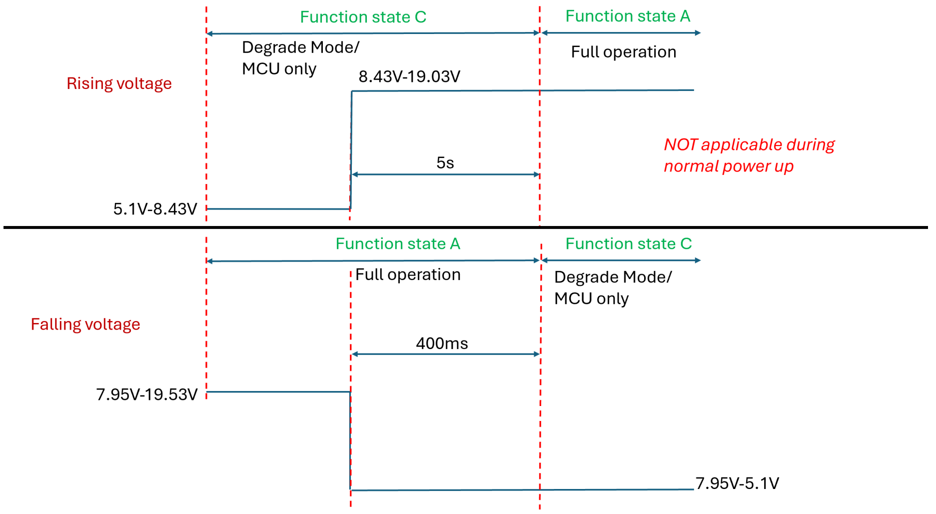
SM-80491 - HW Modes

HWA-1220 - MCU Only state shall be engaged in the following scenarios:

_ <u>MCU only mode to Full operation: **Low** to high input voltage</u>

VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON OR VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is greater than or equal to **0.41984V (3BBh)** for more than 5s ,device goes to full operation mode.

<u>Full operation to MCU only mode : High to **Low** input voltage</u> VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON AND VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is equal to or less than **0.39419V (381h)** for more than 400ms ,device goes to MCU only mode.



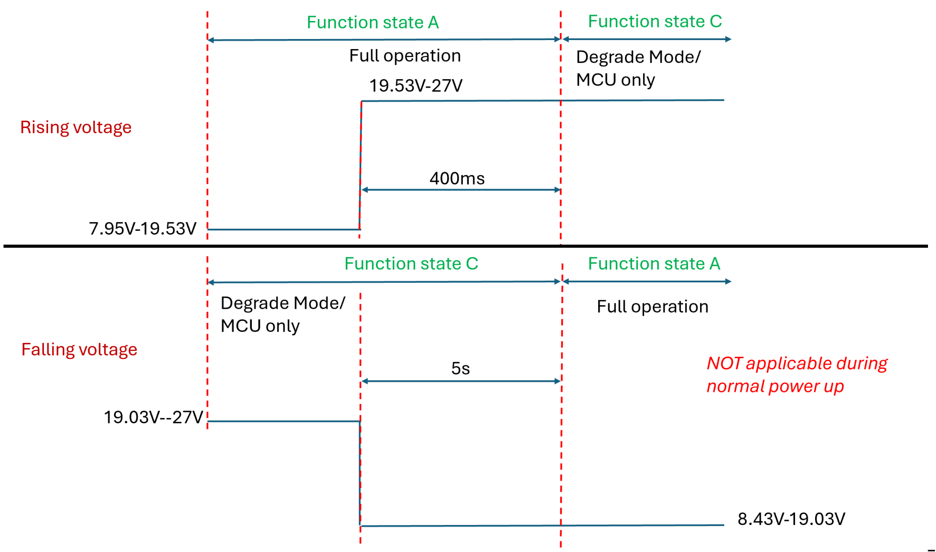
Full operation

MCU only mode : Low to **High** input voltage

VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON AND VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is greater than or equal to **1.00081V (8E5h)** for more than 400ms ,device goes to MCU only mode.

MCU only mode to Full operation : **High** to low input voltage

VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON OR VBAT_INPUT_STAGE.RED_IN_HSS_VMON voltage measured is equal to or less than **0.974118V (8A8h)** for more than 5s ,device goes to full operation mode.



_ [Approved]

4.17.4 VR5510_AMUX Values: HW Safety

SM-80492 - Please refer PLATFORM-HW-DEV-9373 for details [Draft]

SM-71432 -

AMUX [4:0]	Signal Select	P.S Out	Nom.	UV	OV	Remarks
00000	GND	NA	NA	NA	NA	

AMUX [4:0]	Signal Select	P.S Out	Nom.	UV	OV	Remarks
00001	VDDIO voltage divided by 2	3.3V	1.65	1.5847	1.7162	
00010	AMUX Temperature Sensor		0.69 (25°C)	0.495 (125°C)	0.816 (-40°C)	AMUX temp. T(°C) = [(VAMUX - VTEMP25) /VTEMP_COEFF] + 25 > VAMUX=VSUP/20 > VTEMP25=Temp. sensor voltage at 25°C (min. 0.67; typ. 0.69; max. 0.71) > VTEMP_COEFF (min. -2; max. -1.9)[mV/°C]
00011	Bandgap Main	NA	NA	NA	NA	
00100	Bandgap F.S	NA	NA	NA	NA	
00101	BUCK1 voltage	MCU_1v8	1.8	1.756	1.844	
00110	BUCK2 voltage	1.1V	1.1	1.0755	1.1245	
00111	BUCK3 voltage divided by 2.5	MCU_1v1	0.44	0.4191	0.4611	
01000	VPRE voltage divided by 4	MCU_3v3	0.825	0.7722	0.8788	
01001	BOOST Voltage divided by 4	5V_CAN	1.25	1.159	1.3438	
01010	LDO1 voltage divided by 4	MCU_1v8_ANA	0.45	0.4145	0.4862	
01011	LDO2 voltage divided by 4	LDO2_1V8	0.45	0.4145	0.4862	
01100	BOS voltage divided by 4	5V	1.25	0.7841	1.3717	
01101	Reserved					
01110	VSUP1 voltage divided by 20	Vbat_FILT_MCU	0.6	0.2497	1.3870	0.6 is for 12V nominal
01111	PWRON1 voltage divided by 20	PWRON1	0.6	0.2497	1.3870	0.6 is for 12V nominal
10000	PWRON2 voltage divided by 4	NA	NA	NA	NA	
10001	HVLDO voltage divided by 2	MCU_0v8_STBY	0.4	0.3761	0.4241	
10010	LDO3 voltage divided by 4	MCU_LDO3_3v3	0.8	0.7682	0.8831	
Others	GND	NA	NA	NA	NA	

[✔ Approved]

4.18 Voltage Monitoring Validation - HW Safety

4.18.1 TPS389008-Q1 BIST

SM-80493 - TPS389008-Q1 BIST

When the TPS38900x-Q1 is powered ON, BIST is optionally executed (depending on TEST_CFG.AT_POR register bit)
Built-In Self Test (BIST) is performed:

- At Power On Reset (POR), if TEST_CFG.AT_POR=1
- When exiting ACTIVE state due to ACT transitioning from 1→0, if TEST_CFG.AT_SHDN=1

Configuration load from OTP is assisted by ECC (supporting SEC-DED). This is to protect against data integrity issues and to maximize system availability.

TEST_CFG Register (Address = 0x12)

Bit	Field	Type	Default	Description
7:4	RSVD		X	RSVD
3	AT_SHDN		X	Run BIST when exiting ACTIVE state due to ACT transitioning 1 to 0. Device ready after tCFG_WB. This bit cannot be set in OTP/NVM. Always defaults to 0 when loading configuration from OTP/NVM.
2	RESERVED	R	X	
1:0	AT_POR		X	Run BIST at POR. Device ready after tCFG_WB. 00b = Valid OTP configuration, skip BIST at POR 01b = Corrupt OTP configuration, run BIST at POR 10b = Corrupt OTP configuration, run BIST at POR 11b = Valid OTP configuration, run BIST at POR

During BIST, NIRQ is de-asserted (asserted in case of failure), input pins are ignored, and the I2C block is inactive with SDA and SCL de-asserted. The BIST includes device testing to meet the Technical Safety Requirements. Once BIST is completed without failure, I2C is immediately active and the device enters the IDL sate after loading the configuration data from OTP. If BIST fails and/or ECC reports Double-Error Detection (DED), NIRQ is asserted, the device enters FAILSAFE state, and a best effort attempt is made to active I2C. TEST_INFO register may provide additional information on the test results.

The detailed behaviour upon success/failure of the BIST is controlled by INT_TEST and IEN_TEST registers. Reporting of the BIST results is carried out through:

- NIRQ pin: pulled low depending on the test result and BIST_C and BIST bits in IEN_TEST
- I_BIST_C and BIST bits in INT_TEST register depending on IEN_TEST settings
- VMON_STAT.ST_BIST_C register bit
- TEST_INFO[3:0] register bits

TEST_INFO Register (Address = 0x31)

Bit	Field	Type	Default	Description
7:6	RSVD	R	X	RSVD
5	ECC_SEC	R	X	Status of ECC single-error correction on OTP configuration load. 0 = no error correction applied 1 = single-error correction applied
4	ECC_DED	R	X	Status of ECC double-error detection on OTP configuration load. 0 = no double-error detected 1 = double-error detected
3	BIST_VM	R	X	Status of Volatile Memory test output from BIST. 0 = Volatile Memory test pass 1 = Volatile Memory test fail
2	BIST_NVM	R	X	Status of Non-Volatile Memory test output from BIST. 0 = Non-Volatile Memory test pass 1 = Non-Volatile Memory test fail
1	BIST_L	R	X	Status of Logic test output from BIST. 0 = Logic test pass 1 = Logic test fail

Bit	Field	Type	Default	Description
0	BIST_A	R	X	Status of Analog test output from BIST. 0 = Analog test pass 1 = Analog test fail

IEN_TEST Register (Address = 0x1C)

Bit	Field	Type	Default	Description
07:04	RSVD		X	RSVD
3	ECC_SEC		0b	ECC single-error correction fault (on OTP load) interrupt enable: 0 = Interrupt disabled 1 = Interrupt enabled
2	RSVD		X	RSVD
1	BIST_Complete_INT		0b	Built-In Self-Test complete interrupt enable: 0 = Interrupt disabled 1 = Interrupt enabled
0	BIST_Fail_INT		0b	Built-In Self-Test fault interrupt enable: 0 = Interrupt disabled 1 = Interrupt enabled Although expected to be always enabled, it is desirable to have the option to disable it.

INT_TEST Register (Address = 0x23)

Bit	Field	Type	Default	Description
7:4	RSVD	R/W1C	X	RSVD
3	ECC_SEC	R/W1C	X	ECC single-error corrected on OTP configuration load: Write- 1-to-clear will clear the bit. The bit will be set again during next OTP configuration load if the same fault is detected. 0b = No single-error corrected (or IEN_TEST.ECC_SEC is disabled) 1b = Single-error corrected
2	ECC_DED	R/W1C	X	ECC double-error detected on OTP configuration load: The fault bit is always enabled (there is no associated interrupt enable bit). Write- 1-to-clear will clear the bit. The bit will be set again during next OTP configuration load if the same fault is detected. 0b = No double-error detected on OTP load 1b = Double-error detected on OTP load
1	I_BIST_C	R/W1C	X	Indication of Built-In Self-Test complete: Write- 1-to-clear will clear the bit. The bit will be set again on completion of next BIST execution.Write- 1-to-clear will clear the bit. The bit will be set again on completion of next BIST execution. 0b = BIST not complete (or IEN_TEST.BIST_C is disabled) 1b = BIST complete
0	BIST	R/W1C	X	Built-In Self-Test fault: Write- 1-to-clear will clear the bit. The bit will be set again during next BIST execution if the same fault is detected. 0b = No BIST fault detected (or IEN_TEST.BIST is disabled) 1b = BIST fault detected

[✔ Approved]

SM-80494 - TPS389008-Q1 can be used to set register with over voltage and under voltage value. The TPS389008-Q1 is designed to assert active low output signals (NIRQ) when the monitored voltage is outside the safe window.

- Overvoltage threshold can be set using OV_LF[i], INT_OVLF Register (Address = 0x18) detect OV fault has been occurred.
- Undervoltage threshold can be set using UV_LF[i], INT_UVLF Register (Address = 0x14) detect UV fault has been occurred.
- i = 1 to 8

TPS389008-Q1 can be used to set register with over voltage and under voltage value. The TPS389008-Q1 is designed to assert active low output signals (NIRQ) when the monitored voltage is outside the safe window.

- Overvoltage threshold can be set using OV_HF[i], INT_OVHF Register (Address = 0x16) detect OV fault has been occurred.
- Undervoltage threshold can be set using UV_HF[i], INT_UVHF Register (Address = 0x12) detect UV fault has been occurred.
- i = 1 to 8

Details can found in PLATFORM-HW-DEV-9413 and PLATFORM-HW-DEV-14930 [Approved]

SM-80495 - Equation for Analog to digital conversion is:

- Max Voltage below 1.475: [((voltage) -0.2V)/0.005] to hex
- Max Voltage above 1.475: [((voltage) -0.8V)/0.020] to hex

[Approved]

Configuration files of the voltage monitors is in the below link
VM config files

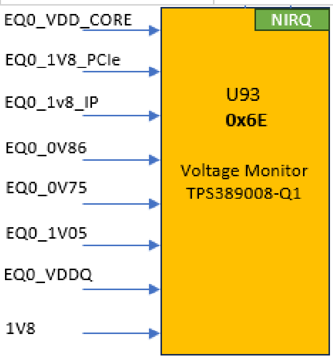
4.18.2 VM#1

4.18.2.1 U93

For configuration file for U93 TPS389008M70RTERQ1 , Check VM config files

SM-71423 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)	Link to OTP
U93 TPS389008M70RTERQ1	EyeQ6H Main	0	0011 0111b	0x37	TPS389008M70RTERQ1_OTP



[Approved]

4.18.2.2 U93 rails list

SM-80497 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 1: 0.875V (core) +2.5% / -3%	EQ0_VDD_CORE	0.84875	0.875	0.896875	☐ PLATFORM-HW-DEV-9386
Chan 2: 1.8V (PCIE) +/-5%	EQ0_1V8_PCIe	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9388
Chan 3: 1.8V (IP) +/-3%	EQ0_1v8_IP	1.746	1.8	1.854	☐ PLATFORM-HW-DEV-9391
Chan 4: 0.86V (PCIE) +/-5%	EQ0_0V86	0.817	0.86	0.903	☐ PLATFORM-HW-DEV-9393
Chan 5: 0.75V (MIPI) +/-5%	EQ0_0V75	0.7125	0.75	0.7875	☐ PLATFORM-HW-DEV-9395
Chan 6: 1.05V (DDR) +/-3%	EQ0_1V05	1.0185	1.05	1.0815	☐ PLATFORM-HW-DEV-9374
Chan 7: 0.5V (DDR) +/-5%	EQ0_VDDQ	0.475	0.5	0.525	☐ PLATFORM-HW-DEV-9375
Chan 8: 1.8V (Main) +/-5%	1V8	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9376

[✔ Approved]

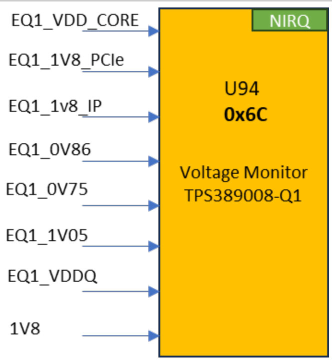
4.18.3 VM#2

4.18.3.1 U94

For configuration file for U94 TPS389008M70RTERQ1 , Check VM config files

SM-71424 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)	Link to OTP
U94 TPS389008M70RTERQ1	EyeQ6H Surround	0	0011 0110b	0x36	TPS389008M70RTERQ1_OTP



[✔ Approved]

4.18.3.2 U94 rails list

SM-80498 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 1: 0.875V (core) +2.5% / -3%	EQ1_VDD_CORE	0.84875	0.875	0.896875	☐ PLATFORM-HW-DEV-9416
Chan 2: 1.8V (PCIE) +/-5%	EQ1_1V8_PCIe	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9417
Chan 3: 1.8V (IP) +/-3%	EQ1_1v8_IP	1.746	1.8	1.854	☐ PLATFORM-HW-DEV-9418

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 4: 0.86V (PCIE) +/-5%	EQ1_0V86	0.817	0.86	0.903	☐ PLATFORM-HW-DEV-9419
Chan 5: 0.75V (MIPI) +/-5%	EQ1_0V75	0.7125	0.75	0.7875	☐ PLATFORM-HW-DEV-9420
Chan 6: 1.05V (DDR) +/-3%	EQ1_1V05	1.0185	1.05	1.0815	☐ PLATFORM-HW-DEV-9421
Chan 7: 0.5V (DDR) +/-5%	EQ1_VDDQ	0.475	0.5	0.525	☐ PLATFORM-HW-DEV-9414
Chan 8: 1.8V (Main) +/-5%	1V8	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9415

[✔ Approved]

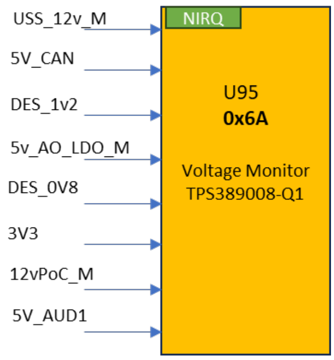
4.18.4 VM#3

4.18.4.1 U95

For configuration file for U95 TPS389008M66RTERQ1, Check VM config files

SM-71427 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit)	Link to OTP
U95 TPS389008M66RTERQ1	Deserializer, CAN, POC, USS & General	0	0011 0101b	0x35	TPS389008M66RTERQ1_OTP



[✔ Approved]

4.18.4.2 U95 rails list

SM-71426 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG	Notes
Chan 1: 1.1V (USS_12v_M) +/-5%	12vPoC_M	1.03835	1.1	1.14765	-	Formula for actual result: Vin x 11
Chan 2: 5V (5V_CAN) +/-5%	5V_CAN	4.75	5	5.25	☐ PLATFORM-HW-DEV-9406	
Chan 3: 1.2V (DES_1v2) +/-5%	DES_1V2	1.14	1.2	1.26	☐ PLATFORM-HW-DEV-9407	
Chan 4: 1.2V (5V_AOLDO_M) +/-5%	5v_AO_LDO_M	1.12575	1.185	1.24425	☐ PLATFORM-HW-DEV-9408	Formula for actual result: Vin x 4.22
Chan 5: 0.8V (DES_0V8) +/-5%	DES_0V8	0.76	0.9	0.84	☐ PLATFORM-HW-DEV-9409	
Chan 6: 3.3V (3V3) +/-5%	3V3	3.135	3.3	3.465	☐ PLATFORM-HW-DEV-9410	

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG	Notes
Chan 7: 1.1V (12vPoC_M) +/-5%	12vPoC_M	1.03835	1.1	1.14765	☐ PLATFORM-HW-DEV-9411	Formula for actual result:Vin x 11
Chan 8: 5V	5V_AUD1	4.75	5	5.25	☐ PLATFORM-HW-DEV-12886	

[✔ Approved]

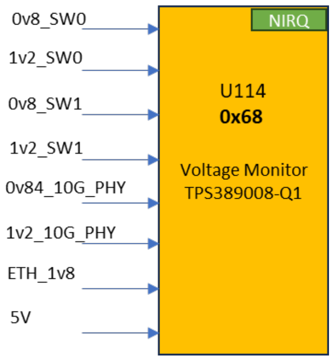
4.18.5 VM#4

4.18.5.1 U114

For configuration file for U114 TPS389008M67RTERQ1, Check VM config files

SM-71428 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)	Link to OTP
U114 TPS389008M67RTERQ1	Switch, Phy& General	0	0011 0100b	0x34	TPS389008M67RTERQ1_OTP



[✔ Approved]

4.18.5.2 U114 rails list

SM-80501 -

VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG	Notes
Chan 1: 0.8V () +/-5%	0v8_SW0	0.76	0.8	0.84	☐ PLATFORM-HW-DEV-9380	
Chan 2: 1.2V () +/-5%	1v2_SW0	1.14	1.2	1.26	☐ PLATFORM-HW-DEV-9378	
Chan 3: 0.8V () +/-5%	0v8_SW1	0.76	0.8	0.84	☐ PLATFORM-HW-DEV-9384	
Chan 4: 1.2V () +/-5%	1v2_SW1	1.14	1.2	1.26	☐ PLATFORM-HW-DEV-9382	
Chan 5: 0.84V () +/-5%	0v84_10G_PHY	0.798	0.84	0.882	☐ PLATFORM-HW-DEV-9389	NA for production
Chan 6: 1.2V () +/-5%	1v2_10G_PHY	1.14	1.2	1.26	☐ PLATFORM-HW-DEV-14228	NA for production
Chan 7: 1.8V (ETH_1v8) +/-5%	ETH_1v8	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9392	
Chan 8: 5V (Main 5V) +/-5%	5V	4.75	5	5.25	☐ PLATFORM-HW-DEV-9390	

[✔ Approved]

4.18.6 VM#5

4.18.6.1 U148

For configuration file for U148 TPS389008M64RTERQ1, Check VM config files

SM-71429 -

Voltage Monitor #	Description	I2C Bus #	Address (binary, 7bit format)	Address (hex, 7bit format)	Link to OTP
U148 TPS389008M64RTERQ1	Encoder	0	0011 0011b	0x33	TPS389008M64_OTP

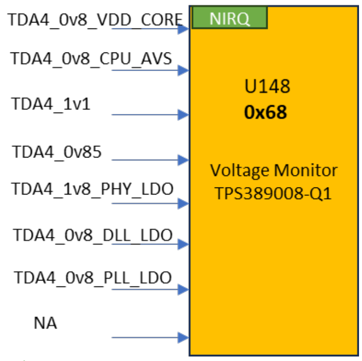
Read registers 0x1B and 0x1C.

If:

0x1B= 0x00 and 0x1C = 0x01

Then update :

0x1B= 0x15 and 0x1C = 0x09



[✔ Approved]

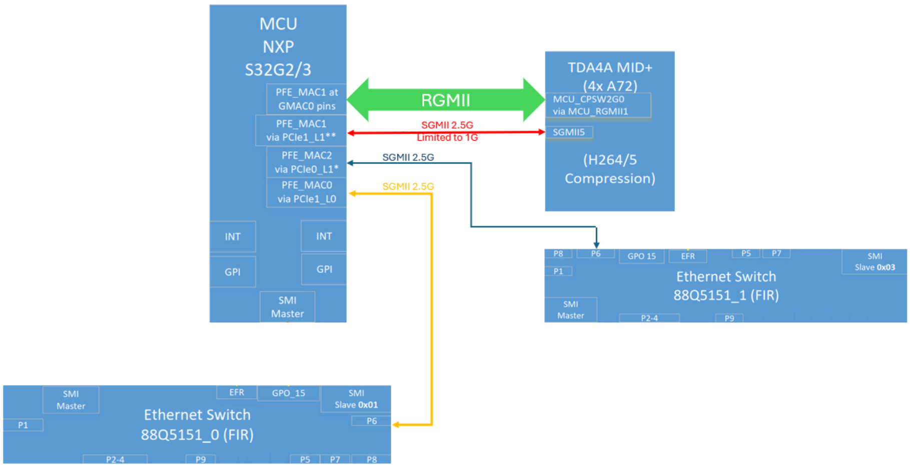
4.18.6.2 U148 rails list

SM-71430 -

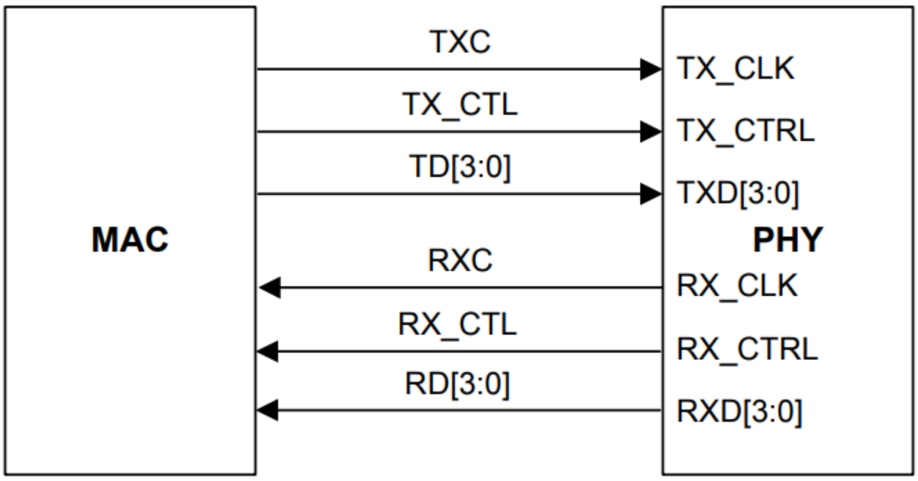
VM Channel	Power Rail	Min.	Typ.	Max.	HWDIAG
Chan 1: 0.8V (TDA4_0v8_VDD_CORE) +/-5%	TDA4_0v8_VDD_CORE	0.76	0.8	0.84	☐ PLATFORM-HW-DEV-9396
Chan 2: 0.8V (TDA4_0v8_CPU_AVS) +/-5%	TDA4_0v8_CPU_AVS	0.76	0.8	0.84	☐ PLATFORM-HW-DEV-9394
Chan 3: 1.1V (TDA4_1v1) +/-5%	TDA4_1v1	1.045	1.1	1.155	☐ PLATFORM-HW-DEV-9377
Chan 4: 0.85V (TDA4_0v85) +/-5%	TDA4_0v85	0.8075	0.85	0.8925	☐ PLATFORM-HW-DEV-9381
Chan 5: 1.8V (TDA4_1v8_PHY_LDO) +/-5%	TDA4_1v8_PHY_LDO	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9379
Chan 6: 0.8V (TDA4_0v8_DLL_LDO) +/-5%	TDA4_0v8_DLL_LDO	0.76	0.8	0.84	☐ PLATFORM-HW-DEV-9385
Chan 7: 1.8V (TDA4_1v8_PLL_LDO) +/-5%	TDA4_1v8_PLL_LDO	1.71	1.8	1.89	☐ PLATFORM-HW-DEV-9383
Chan 8: 0V disabled	P.D		0		

[✔ Approved]

4.19 MCU Ethernet Connectivity



4.20 MCU – TDA4 Main Link: RGMII



MCU – PFE_MAC1 **TDA4 (CPSW2G0)**

4.20.1 RGMII signal connectivity

Data signals are working at 150MHz

Pin#	Peripheral	Pin/Port#	Net Name
P20	PFE_MAC1	[P20] PE_02	ETH.ENC_3v3_RGMII1_RX_CLK
U23	PFE_MAC1	[U23] PE_03	ETH.ENC_3v3_RGMII1_RX_CTL
T23	PFE_MAC1	[T23] PE_04	ETH.ENC_3v3_RGMII1_RD0
T22	PFE_MAC1	[T22] PE_05	ETH.ENC_3v3_RGMII1_RD1
U22	PFE_MAC1	[U22] PE_06	ETH.ENC_3v3_RGMII1_RD2
T21	PFE_MAC1	[T21] PE_07	ETH.ENC_3v3_RGMII1_RD3
R21	PFE_MAC1	[R21] PE_08	ETH.ENC_3v3_RGMII1_TX_CLK
R23	PFE_MAC1	[R23] PE_09	ETH.ENC_3v3_RGMII1_TX_CTL

Pin#	Peripheral	Pin/Port#	Net Name
P19	PFE_MAC1	[P19] PE_10	ETH.ENC_3v3_RGMII1_TX_D0
P18	PFE_MAC1	[P18] PE_11	ETH.ENC_3v3_RGMII1_TX_D1
N18	PFE_MAC1	[N18] PE_12	ETH.ENC_3v3_RGMII1_TX_D2
U21	PFE_MAC1	[U21] PE_13	ETH.ENC_3v3_RGMII1_TX_D3

4.21 MCU SGMII

4.21.1 Chosen working modes

SM-80505 - ETH.MCU_2G5_SGMII should be configured as below
serdes0:mode=xpcs0&xpcs1,clock=int,fmhz=100;xpcs0_0:speed=2G5;xpcs0_1:speed=2G5 **[** ✓ Approved **]**

SM-80506 - ETH.MCU_2G5_SGMII_1 should be configured as below
serdes1:mode=xpcs0&xpcs1,clock=int,fmhz=125;xpcs1_0:speed=2G5;xpcs1_1:speed=2G5 **[** ✓ Approved **]**

4.21.2 SGMII Connectivity

SM-80507 -

SGMII#	Mode	PHY Lane 0	PHY Lane 1	Connect to	ext. Ref. Clock
0	SGMII Only		PFE_MAC2	ETH. SWITCH_0	NA
1	SGMII 2.5G	PFE_MAC0		ETH. SWITCH_1	NA

[✓ Approved **]**

* Note: MCU - TDA4 SGMII link is not used

4.22 I2C

4.22.1 Buses

#	Peripheral	Signal	Routed pin/signal	Label
W12	I2C_0	sda, 0	PB_00	MCU_IF.3v3_I2C0_SDA
E7	I2C_0	scl, 0	PB_01	MCU_IF.3v3_I2C0_SCL
A6	I2C_2	scl, 2	PB_05	MCU_IF.3v3_I2C2_SCL
G9	I2C_2	sda, 2	PB_06	MCU_IF.3v3_I2C2_SDA
Y8	I2C_4	sda, 4	PC_01	MCU_IF.3v3_I2C4_SDA
W8	I2C_4	scl, 4	PC_02	MCU_IF.3v3_I2C4_SCL
C6	I2C_1	scl, 1	PB_03	MCU_IF.3v3_I2C1_SCL
E8	I2C_1	sda, 1	PB_04	MCU_IF.3v3_I2C1_SDA

4.22.2 Addresses

SM-72357 -

Device	Ref. Des.	I2C Bus #	Add. (7 bit)	Comments
PMIC BUCK	U50	0	0x28	
PMIC	U54	4	Main = 0x20	
			Safety = 0x21	

Device	Ref. Des.	I2C Bus #	Add. (7 bit)	Comments
GPIO Ex.	U103	0	0x76	
Temp. Sensor	U61	0	0x4C	
V.M	U148	0	0x33	
P.S	U32	2	0x30	
eePROM	U59	0	0x50	
P.S	B_U100	2	0x46	
Temp. Sensor	U60	0	0x48	
Temp. Sensor	U62	0	0x49	
P.S	B_U26	2	0x40	
Temp. Sensor	U64	0	0x4A	
P.S	B_U18	2	0x43	
P.S	A_U26	1	0x40	
P.S	A_U18	1	0x43	
P.S	A_U100	1	0x46	
Clk Gen.	U88	1	0x6B	
Clk Gen.	U87	1	0x6A	
P.S	U33	1	0x33	
V.M	U93	0	0x37	
V.M	U94	0	0x36	
V.M	U95	0	0x35	
V.M	U114	0	0x34	
GPIO Ex.	U104	0	0x75	
GPIO Ex.	U146	0	0x74	
P.S	U34	2	0x43	
USS P.S	U91	1	0x75	N.U

[Approved]

4.23 Temperature Sensors - HW Safety

SM-80510 - The thermal reaction is defined below

1. Critical Temperature high (low) Temperature (per SYS.3):
   a. Operating beyond this temperature shall be avoided
   b. there's time left to do graceful shutdown considering possible temperature gradients
   c. Perform graceful shutdown
2. Non-Critical high (Low) Temperature (SYS.3):
   a. operating outside specified temperature range
   b. just for logging purposes / quality analysis in the field etc

[Draft]

4.23.1 TMP75B-Q1

4.23.1.1 Addresses & Location

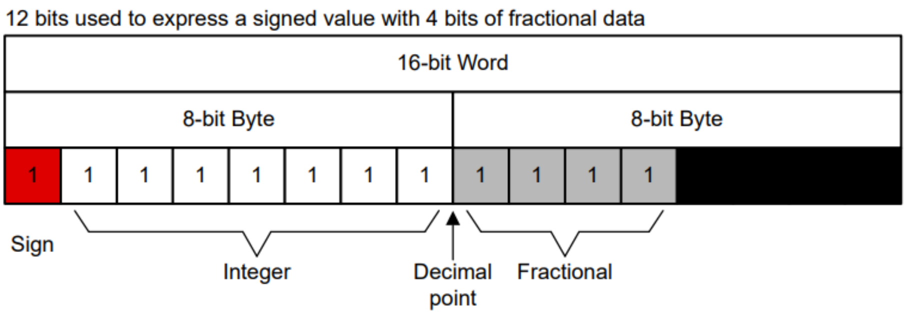
SM-72369 -

Temp. Sense #	Location	I2C Bus #	Add. (hex, 7bit)
U60	Near EyeQ_1 DDR	0	0x48
U61	Near ENCODER DDR	0	0x4C
U62	Between EyeQ_0 & EyeQ_1 DDRs	0	0x49
U64	Between EyeQ_0 & MCU DDRs	0	0x4A

[✔ Approved]

4.23.1.2 Temperature Sensors Read Conversion

SM-80511 - Negative numbers are represented in binary twos complement format. Following power-up or reset, the Temperature register reads 0°C until the first conversion is complete. The user can obtain 9, 10, 11, or 12 bits of resolution by addressing the Configuration register and setting the resolution bits accordingly. For 9-, 10-, or 11-bit resolution, the most significant bits (MSBs) in the Temperature register are used with the unused least significant bits (LSBs) set to zero



[✔ Approved]

SM-80512 - The Configuration register is an 16-bit read/write register used to store bits that control the operational modes of the temperature sensor. Read and write operations are performed MSB first. The power-up or reset value of the Configuration register are all bits equal to 0.

BYTE	D15	D14	D13	D12	D11	D10	D9	D8
1	OS	CR		FQ		POL	TM	SD

BYTE	D7	D6	D5	D4	D3	D2	D1	D0
0	Reserved	Read only -FFh						

Below Configuration can be used for configuring the TMP75B-Q1

D15: D8 - 0b 00000010

Number of consecutive faults: 0

Polarity: 0, Alert pin will be active low

Conversion rate : 37Hz (continuous conversion)

TM:1 Interrupt mode is used

Shutdown mode disabled

[✔ Approved]

For the temperature monitoring registers please refer
Chapter 1.5 of *SuppliersManagement/Primary Board HW Diagnostics/HWDIAG*

4.23.1.3 Temperature limits

SM-80514 -

THIGH register is used to set the higher threshold and **T**LOW register used to set the lower threshold for TMP75-Q1.

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature[c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy	HWDIAG
Temp_sense1 (U60)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐PLATFORM-HW-DEV-12630
Temp_sense5 (U61)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐PLATFORM-HW-DEV-12637
Temp_sense3 (U62)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐PLATFORM-HW-DEV-12633
Temp_sense2 (U127)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐PLATFORM-HW-DEV-12632
Temp_sense4 (U64)	105 0x690	-40 0xD80	98 0x620	-32 0xE00	-20 0xEC0	1	☐PLATFORM-HW-DEV-12635

Cooling TH: Temperature above which cooling is requested **[**✔ Approved **]**

4.23.2 S32G internal temperature sensor

4.23.2.1 Configuration

SM-80515 -

- TMU base address: 400A_8000h
- Mode (TMR) register with Offset: 0h allows the software to control the thermal monitoring operation: Mode, enable/disable, Low pass filter
- Status (TSR) register with Offset: 4h reports the monitoring and calibration status during operation.
- Monitor Site (TMSR) register with Offset: 8h indicates which remote site sensor is actively being monitored by TMU when TMR[MODE]=10b
- Monitor Temperature Measurement Interval (TMTMIR) register with Offset: Ch allows you to control at what frequency the module reads the temperature sensors.
- Report Immediate Temperature at Site (TRITSR0 - TRITSR2) registers indicates Last Temperature Reading in Kelvin at the Site
  - TRITSR0 with Offset: 100h
  - TRITSR1 with Offset: 110h
  - TRITSR2 with Offset: 120h

- TMU interrupt: NCF designation is FCCU NCF [46] and NVIC interrupt ID is 120. Following interrupts provide FCCU interrupt
  - Immediate High Temperature Threshold Interrupt Enable
  - Immediate Low Temperature Threshold Interrupt Enable
  - Average High Temperature Threshold Interrupt Enable
  - Average Low Temperature Threshold Interrupt Enable
- Threshold for the temperature can be set using following registers
  - Monitor High Temperature Immediate Threshold (TMHTITR) with Offset: 50h

- Monitor High Temperature Average Threshold (TMHTATR) with Offset: 54h
- Monitor Low Temperature Immediate Threshold (TMHTITR) with Offset: 60h
- Monitor Low Temperature Average Threshold (TMHTATR) with Offset: 64h

- Temperature Configuration (TTCFGR) with Offset: 80h register, in conjunction with Sensor Configuration (TSCFGR), is used to initialize the internal sensor translation table used during monitoring.
- Sensor Configuration (TSCFGR) with Offset: 84h register, in conjunction with Sensor Configuration (TSCFGR), is used to initialize the internal sensor translation table used during monitoring.
- Interrupt Enable (TIER) register with Offset: 20h determines if a detected status condition causes a system interrupt.
- Interrupt Detect (TIDR) register with Offset: 24h indicates if a status condition is detected that could generate an interrupt.
- Interrupt Average Site Capture (TIASCR) register with Offset: 34h indicates which remote sensor sites are associated with the detection of an interrupt event related to average reported temperatures.
- Interrupt Immediate Site Capture (TIISCR) register with Offset: 30h indicates which remote sensor sites are associated with the detection of an interrupt event related to current reported temperatures.

The procedure to enable the monitoring mode is as follows:

1. Clear the interrupt detect register, Interrupt Detect (TIDR)
2. Clear the interrupt capture registers: Interrupt Immediate Site Capture (TIISCR) and Interrupt Average Site Capture (TIASCR)
3. Enable interrupt handling by writing 1 to the appropriate fields in the interrupt enable register, Interrupt Enable (TIER).
4. Write the appropriate values to the following temperature threshold registers:
   • Monitor High Temperature Immediate Threshold (TMHTITR)
   • Monitor Low Temperature Immediate Threshold (TMLTITR)
   • Monitor High Temperature Average Threshold (TMHTATR)
   • Monitor Low Temperature Average Threshold (TMLTATR)
5. Write the appropriate value to the monitoring interval register, Monitor Temperature Measurement Interval (TMTMIR).
6. Select the ALPF coefficient in the Mode (TMR). The ALPF coefficient is used in the average temperature calculations.
7. Enable the monitor mode by setting TMR[MODE] to 10b. You control the monitored sites by writing an appropriate value to TMSR[SITE]. You must enable at least one active site, and set the other mode control fields as needed.
8. If the monitoring interval is too short, as indicated by TSR[MIE], you may need to increase the interval or reduce the number of sites.

[Approved **]**

SM-80516 - S32G threshold configuration

The below registers are configured with respect to temperature limit table

Monitor High Temperature Immediate Threshold (TMHTITR): Critical Temperature High [c]

Monitor Low Temperature Immediate Threshold (TMLTITR): Critical Temperature Low [c]

Monitor High Temperature Average Threshold (TMHTATR): Non-Critical High Temperature[c]

Monitor Low Temperature Average Threshold (TMLTATR): Non-Critical Low Temperature[c] **[** In Review **]**

4.23.2.2 Temperature limits

SM-80517 -

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature[c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy
MCU_Temp_sens_0	125 0x18E	-40 0x0E9	120 0x189	-36 0x0ED	-20 0x0FD	
MCU_Temp_sens_1	125 0x18E	-40 0x0E9	120 0x189	-36 0x0ED	-20 0x0FD	

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature[c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy
MCU_Temp_sens_2	125 0x18E	-40 0x0E9	120 0x189	-36 0x0ED	-20 0x0FD	

NOTE: temperature conversion is done by converting temperature to Kelvin to a 9bit hex [✔ Approved]

4.23.3 EyeQ internal temperature sensor

4.23.3.1 Configuration

EyeQ6H Thermal Application Note

4.23.3.2 Temperature limits

SM-80519 - EyeQ and TDA4 Limits

Name	Critical Temperature High [c]	Critical Temperature Low [c]	Non-Critical High Temperature [c]	Non-Critical Low Temperature[c]	Cooling TH	Accuracy
EQ_0_Temp_sens_Combined	120	-35	118	-30	NA	3.5
EQ_1_Temp_sens_Combined	120	-35	118	-30	NA	3.5
TDA4_Temp_sens_Combined	125	-35	118	-30	NA	

[✔ Approved]

4.24 SMI

4.24.1 MDC/IO port

SM-80520 -

Peripheral	Signal	Pin/Port	Label
PFE_MAC0	mdc	[M19] PF_00	MCU_IF.1v8_MDC
PFE_MAC0	md	[L22] PF_01	MCU_IF.1v8_MDIO

[✔ Approved]

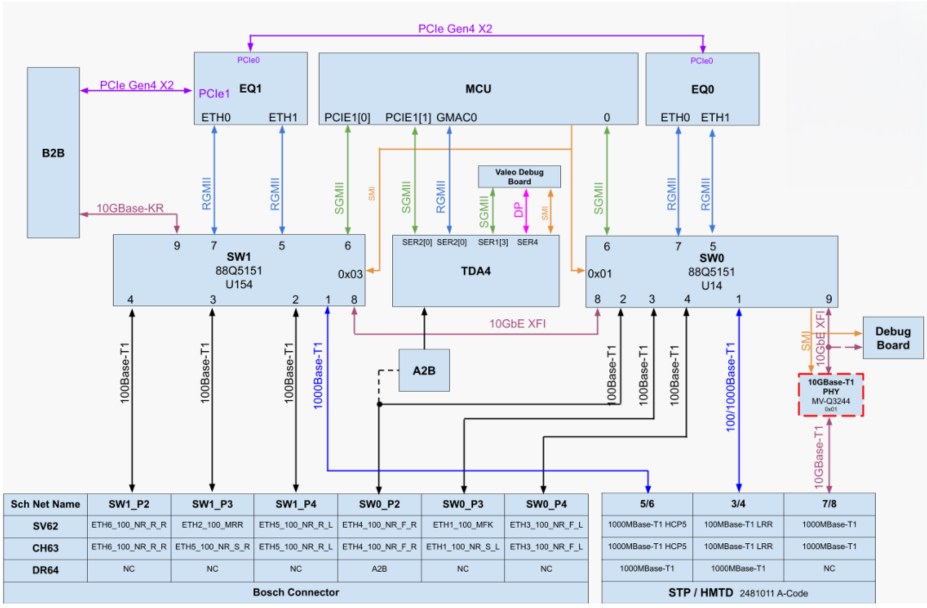
4.24.2 SMI Address List

SM-72374 -

Device	P.N	Ref. Des.	Address	Notes
ETH. SW_0	88Q5151	U14	0x01	
ETH. SW_1	88Q5151	U154	0x03	

[✔ Approved]

4.25 Ethernet Connectivity



4.25.1 Ethernet Porting List

4.25.1.1 SW0 – U14

SM-72379 -

Port	Interface	Connect to
1	1000BASE-T1	HMTD Conn.
2	N.C	N.C
3	N.C	N.C
4	N.C	N.C
5	RGMII	EQ6H_0 Eth1
6	2.5G	MCU PCIe0_L1
7	RGMII	EQ6H_0 Eth0
8	10G	SW1 P8
9	N.C	N.C

[✅ Approved]

4.25.1.2 SW1– U154

SM-72380 -

Port	Interface	Connect to
1	1000BASE-T1	HMTD Conn.
2	N.C	N.C
3	N.C	N.C

Port	Interface	Connect to
4	N.C	N.C
5	RGMII	EQ6H_1 Eth0
6	2.5G	MCU PCIe1_L0
7	RGMII	EQ6H_1 Eth1
8	10G	SW0 P8
9	10G	B2B

[✔ Approved]

4.26 Interrupts - HW Safety

SM-72383 -

Sources	EIRQ	Port	Trigger Type	Net Name	Yes/No
ETH. SW0	EIRQ[20]	PL_12	Falling	SAFETY_SIG.SW0_1V8_EFR	Yes
	EIRQ[22]	PL_14		SAFETY_SIG.SW0_1V8_HARTBEAT	Yes
ETH. SW1	EIRQ[21]	PL_13	Falling	SAFETY_SIG.SW1_1V8_EFR	Yes
	EIRQ[23]	PH_01		SAFETY_SIG.SW1_3V3_HARTBEAT	Yes
EyeQ main	EIRQ[4]	PL_03	Rising+Falling	SAFETY_SIG.EQ0_3v3_CRERR	Yes
	EIRQ[5]	PB_08	Rising+Falling	SAFETY_SIG.EQ0_3v3_NCRERR	Yes
CAN	EIRQ[16]	PL_08	Falling	CAN.3v3_RADAR4_ERR	NA
	EIRQ[17]	PL_09	Falling	CAN.3v3_RADAR5_ERR	NA
EyeQ Surround	EIRQ[12]	PB_15	Rising+Falling	SAFETY_SIG.EQ1_3v3_CRERR	Yes
	EIRQ[15]	PC_03	Rising+Falling	SAFETY_SIG.EQ1_3v3_NCRERR	Yes
Temp. Sensor	EIRQ[27]	PC_04	Rising+Falling	TEMP_SENS_ALERT	Yes
Voltage Monitors	EIRQ[11]	PK_08	Rising+Falling	SAFETY_SIG.VMs_3v3_INT	Yes
VBAT Input	EIRQ[1]	PK_05	Rising+Falling	VBAT_INPUT_STAGE.3v3_DIODE_RED_PG	Yes
	EIRQ[26]	PH_05	Rising+Falling	VBAT_INPUT_STAGE.3V3_MAIN_HSS_STATUS	Yes
	EIRQ[24]	PH_02	Rising+Falling	SAFETY_SIG.3v3_Main_BAT_Fault_INT	Yes
	EIRQ[7]	PB_10	Rising+Falling	SAFETY_SIG.3v3_Red_BAT_Fault_INT	Yes
	EIRQ[25]	PH_03	Rising+Falling	VBAT_INPUT_STAGE.3v3_DIODE_PG	Yes
MCU PMIC	EIRQ[31]	PC_08	Falling	PMIC_INTB	Yes
TDA4	EIRQ[19]	PL_11		ENC.1v8_MCU_TIMESYNC1	NA
	EIRQ[2]	PL_01	Rising+Falling	SAFETY_SIG.3v3_ENC_PMIC_SS	Yes
	EIRQ[24]	PD_15	Falling	ENC.1v8_MCU_PORZ	Yes
	EIRQ[3]	PL_02	Rising+Falling	MCU_CTL.3v3_ENC_PMIC_RST_OUT	Yes

No: Polling is used

Yes: Interrupt is used

NA: Will not be monitored [✔ Approved]

4.27 GPIO init state

SM-80522 - Production vs Debug GPIO init

- When debug level =0, then the production GPIO init configuration needs to be applied
- When debug level =any other value, then the debug GPIO init configuration needs to be applied

SM-3778 - debug level
[Approved]

4.27.1 MCU GPIOs init state

4.27.1.1 GPIOs init state

4.27.1.1.1 Debug

SM-72387 - MCU GPIOs init state -Debug

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[W10] PA_02	BOOTMOD0	BOOT	Input	<NA>	PD			50MHz 3.3V
[W11] PA_03	BOOTMOD1	BOOT	Input	<NA>	PD			50MHz 3.3V
[Y11] PA_07	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[U10] PA_08	EQ_IOs.EQ0_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[B8] PA_09	EQ_IOs.EQ1_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[E13] PA_10	DEBUG.3v3_DETECT	GPIO	Input	<NA>				50MHz 3.3V
[D8] PA_11	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C7] PA_12	CAN.3v3_B2B_EN	GPIO	Output	Low				50MHz 3.3V
[U12] PA_13	N.U	GPIO	High-Z	<NA>	PD			1MHz 3.3V
[W13] PA_15	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[W12] PB_00	MCU_IF.3v3_I2C0_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[E7] PB_01	MCU_IF.3v3_I2C0_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[D7] PB_02	WAKE.3v3_MCU_KEEPALIVE	GPIO	Output	Low				50MHz 3.3V
[AA10] PB_09	CAN.3v3_B2B_TX	CAN	Output	<NA>			500Kbps	50MHz 3.3V
[E6] PB_14	CAN.3v3_B2B_RX	CAN	Input	<NA>			500Kbps	50MHz 3.3V
[A6] PB_05	MCU_IF.3v3_I2C2_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[G9] PB_06	MCU_IF.3v3_I2C2_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[F7] PB_08	SAFETY_SIG.EQ0_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[U18] PH_02	SAFETY_SIG.3v3_Main_BAT_Fault_INT	EIRQ	Input	<NA>	PD			83MHz 3.3V
[V11] PB_10	SAFETY_SIG.3v3_Red_BAT_Fault_INT	EIRQ	Input	<NA>	PD			50MHz 3.3V
[G8] PB_11	MCU_IF.FR_TXEN	GPIO	Output	Low				50MHz 3.3V
[F8] PB_12	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[G7] PB_13	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y9] PA_06	EFUSE_EN	GPIO	Output	Low				50MHz 3.3V
[B5] PB_15	SAFETY_SIG.EQ1_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[V9] PC_00	DISCHARGE.ETH_MCU	GPIO	Output	Low				50MHz 3.3V
[Y8] PC_01	MCU_IF.3v3_I2C4_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[W8] PC_02	MCU_IF.3v3_I2C4_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[U9] PC_03	SAFETY_SIG.EQ1_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[C13] PC_04	TEMP_SENS_ALERT	EIRQ	Input	<NA>				50MHz 3.3V
[G10] PC_05	ENC.3v3_UART_RX_R	GPIO	Input	<NA>			115Kbps	50MHz 3.3V
[A14] PC_06	ENC.3v3_UART_TX_R	GPIO	Input	<NA>				50MHz 3.3V
[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[B6] PC_08	PMIC_INTB	EIRQ	Input	<NA>				50MHz 3.3V
[U11] PC_09	MCU_IF.3v3_MCU_CLI_UART_TX	UART	Input	<NA>			115Kbps	50MHz 3.3V
[Y12] PC_10	MCU_IF.3v3_MCU_CLI_UART_RX	UART	Input	<NA>			115Kbps	50MHz 3.3V
[F15] PC_11	CAN.3v3_HCP5_RX	CAN	Input	<NA>			2Mbps	50MHz 3.3V
[G11] PC_12	CAN.3v3_HCP5_TX	CAN	Output	<NA>			2Mbps	50MHz 3.3V
[Y7] PC_13	CAN.3v3_HCP5_EN	GPIO	Input Output	Low				50MHz 3.3V
[Y10] PD_11	EQ_IOs.EQ0_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[V23] PD_12	EQ_IOs.EQ0_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[R19] PD_13	EQ_IOs.EQ1_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[N23] PH_00	EQ_IOs.EQ0_1_1v8_GPIO_B14_SYNC	GPIO	Output	Low				150MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[P23] PD_14	ENC.1v8_PORZ	GPIO	Output	Low				150MHz 1.8V
[P20] PE_02	ETH.ENC_3v3_RGMII1_RX_CLK	GPIO	Output	Low				100MHz 3.3V
[U23] PE_03	ETH.ENC_3v3_RGMII1_RX_CTL	GPIO	Output	Low				150MHz 3.3V
[T23] PE_04	ETH.ENC_3v3_RGMII1_RD0	GPIO	Output	Low				150MHz 3.3V
[T22] PE_05	ETH.ENC_3v3_RGMII1_RD1	GPIO	Output	Low				150MHz 3.3V
[U22] PE_06	ETH.ENC_3v3_RGMII1_RD2	GPIO	Output	Low				150MHz 3.3V
[T21] PE_07	ETH.ENC_3v3_RGMII1_RD3	GPIO	Output	Low				150MHz 3.3V
[R21] PE_08	ETH.ENC_3v3_RGMII1_TX_CLK	RGMII CLK	Input	<NA>				150MHz 3.3V
[R23] PE_09	ETH.ENC_3v3_RGMII1_TX_CTL	RGMII CTL	Input	<NA>				150MHz 3.3V
[P19] PE_10	ETH.ENC_3v3_RGMII1_TX_D0	RGMII DATA	Input	<NA>				150MHz 3.3V
[P18] PE_11	ETH.ENC_3v3_RGMII1_TX_D1	RGMII DATA	Input	<NA>				150MHz 3.3V
[N18] PE_12	ETH.ENC_3v3_RGMII1_TX_D2	RGMII DATA	Input	<NA>				150MHz 3.3V
[U21] PE_13	ETH.ENC_3v3_RGMII1_TX_D3	RGMII DATA	Input	<NA>				150MHz 3.3V
[T18] PE_14	CAN.3v3_RADAR1_STB	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[L22] PF_01	MCU_IF.1v8_MDIO	MDIO	Input Output	<NA>				150MHz 1.8V
[V17] PE_15	CAN.3v3_RADAR5_STB	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[M19] PF_00	MCU_IF.1v8_MDC	MDC	Output	<NA>				150MHz 1.8V
[Y21] PF_02	CAN.3v3_RADAR5_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[AA12] PA_14	SAFETY_SIG.DC_EQs_IN	GPIO	Output	High				50MHz 3.3V
[V14] PF_03	TDA4_EFUSE_EN	GPIO	Output	Low				150MHz 1.8V
[V13] PF_04	ENC.1v8_SoC_BOOTMODE0	GPIO	Output	Low				150MHz 1.8V
[U8] PF_15	MCU_CTL.power_EN_n	GPIO	Output	Low				50MHz 3.3V
[T19] PH_01	SAFETY_SIG.SW1_3v3_HARTBEAT	EIRQ	Input	<NA>				150MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[L21] PD_15	ENC.1v8_MCU_PORZ	EIRQ	Output	Low				150MHz 1.8V
[U20] PH_03	VBAT_INPUT_STAGE.3V3_DIODE_PG	EIRQ	Input	<NA>	PU			150MHz 3V3
[P21] PE_00	EQ_IOs.EQ1_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[V21] PH_04	CAN.3v3_RADAR3_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W23] PH_05	VBAT_INPUT_STAGE.3V3_MAIN_HSS_STATUS	EIRQ	Input	<NA>	PU			150MHz 3.3V
[T20] PH_06	CAN.3v3_RADAR4_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y23] PH_07	CAN.3v3_RADAR4_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W21] PH_08	CAN.3v3_RADAR2_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[M23]PE_01	EQ_IOs.EQ0_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[W22] PH_09	CAN.3v3_RADAR2_STBn	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[V20] PH_10	CAN.3v3_RADAR1_EN	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[U19] PJ_00	CAN.3v3_RADAR3_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[B11] PJ_01	CAN.3v3_RADAR2_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D17] PJ_02	CAN.3v3_RADAR2_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D10] PJ_03	CAN.3v3_RADAR3_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C15] PJ_04	CAN.3v3_RADAR3_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C11] PJ_05	CAN.3v3_RADAR4_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D15] PJ_06	CAN.3v3_RADAR4_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B15] PK_06	SAFETY_SIG.PMIC_3v3_SS	GPIO	Input	<NA>				50MHz 3.3V
[D12] PJ_07	CAN.3v3_RADAR1_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[D11]PK_05	VBAT_INPUT_STAGE.3v3_DIODE_RED_PG	EIRQ	Input	<NA>	PU			50MHz 3.3V
[F16] PJ_08	CAN.3v3_RADAR3_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[B12] PJ_09	MCU_CTL.SW0_RST	GPIO	High-Z	High				50MHz 3.3V
[E10] PK_09	EQ_IOs.3v3_MCU_EQ0_TX	CAN	Output	<NA>				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[E14] PK_10	EQ_IOs.3v3_MCU_EQ0_RX	CAN	Input	<NA>				50MHz 3.3V
[E12] PJ_11	CAN.3v3_DEBUG_TX	CAN	Output	<NA>			500Kbps	50MHz 3.3V
[E16] PJ_12	CAN.3v3_DEBUG_RX	CAN	Input	<NA>			500Kbps	50MHz 3.3V
[A11] PJ_13	CAN.3v3_RADAR5_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C16] PJ_14	CAN.3v3_RADAR5_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[A7] PL_04	CAN.3v3_B2B_STBn	GPIO	Output	Low				50MHz 3.3V
[F11] PK_01	EQ_IOs.3v3_MCU_EQ1_TX	CAN	Output	<NA>				50MHz 3.3V
[D14] PK_02	EQ_IOs.3v3_MCU_EQ1_RX	CAN	Input	<NA>				50MHz 3.3V
[C6] PB_03	MCU_IF.3v3_I2C1_SCL	I2C SCL	Output	<NA>		OD	400KHz	50MHz 3.3V
[F14] PK_04	IGN_SENSE	GPIO	Input	<NA>				50MHz 3.3V
[E8] PB_04	MCU_IF.3v3_I2C1_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[A15] PK_00	CAN.3v3_HCP5_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[C10] PK_07	EQ_IOs.EQ1_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[F13] PK_08	SAFETY_SIG.VMs_3v3_INT	EIRQ	Input	<NA>				50MHz 3.3V
[A10] PJ_15	CAN.3v3_HCP5_STBn	GPIO	Input Output	Low				50MHz 3.3V
[E11] PK_03	HCP5_INH	GPIO	Input	<NA>				50MHz 3.3V
[A9] PK_11	DEBUG.3v3_MCU_UART_TX	UART	Output	<NA>			115Kbps	50MHz 3.3V
[C14] PK_12	DEBUG.3v3_MCU_UART_RX	UART	Input	<NA>			115Kbps	50MHz 3.3V
[C9] PK_13	CAN.3v3_RADAR1_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B13] PK_14	CAN.3v3_RADAR1_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[E9] PL_01	SAFETY_SIG.3v3_ENC_PMIC_SS	EIRQ	Input	<NA>				50MHz 3.3V
[A8] PL_02	MCU_CTL.3v3_ENC_PMIC_RST_OUT	EIRQ	Input	<NA>				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[A13] PL_03	SAFETY_SIG.EQ0_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[D16] PJ_10	CAN.3v3_RADAR2_ERRn	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C8] PL_05	CAN.3v3_B2B_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[D13] PL_06	B2B_CAN_INH	GPIO	Input	<NA>				50MHz 3.3V
[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[N19] PL_08	CAN.1v8_RADAR4_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[A5] PB_07	SAFETY_SIG.DC_EQ1_OUT	GPIO	Input	<NA>				50MHz 3.3V
[L23] PL_09	CAN.1v8_RADAR5_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[P22] PL_10	ENC.1v8_MCU_TIMESYNC0	GPIO	Input	<NA>	PD			166MHz 1.8V
[N22] PL_11	ENC.1v8_MCU_TIMESYNC1	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[N20] PL_12	SAFETY_SIG.SW0_1V8_EFR	EIRQ	Input	<NA>				150MHz 1.8V
[N21] PL_13	SAFETY_SIG.SW1_1V8_EFR	EIRQ	Input	<NA>				150MHz 1.8V
[M21] PL_14	SAFETY_SIG.SW0_1V8_HARTBEAT	EIRQ	Input	<NA>				166MHz 1.8V
[K20] PG_00	QSPI_A_SCK	QSPI CK	Output	<NA>			100MHz	133MHz 1.8V
[K21] PG_01	N.C	QSPI_A_CK	High-Z	<NA>	PD			150MHz 1.8V
[J21] PG_02	N.C	QSPI_A_CK2	High-Z	<NA>	PD			150MHz 1.8V
[H21] PG_03	N.C	QSPI_A_CK2	High-Z	<NA>	PD			150MHz 1.8V
[G21] PG_04	QSPI_A_CS0	QSPI_A_CS0	Output	Low				133MHz 1.8V
[H22] PG_05	TEST_POINT	QSPI_A_CS1	Output	Low				150MHz 1.8V
[L18] PF_05	QSPI_A_DATA0	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L19] PF_06	QSPI_A_DATA1	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L20] PF_07	QSPI_A_DATA2	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K22] PF_08	QSPI_A_DATA3	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[K19] PF_09	QSPI_A_DATA4	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[J23] PF_10	QSPI_A_DATA5	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[H23] PF_11	QSPI_A_DATA6	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K18] PF_12	QSPI_A_DATA7	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K23] PF_13	QSPI_A_DQS	QSPI DQS	Input	<NA>	PD			133MHz 1.8V
[J19] PF_14	QSPI_A_INT	QSPI INT	Input	<NA>				133MHz 1.8V
[E19] PC_14	eMMC_CK	USDHC CK	Output	<NA>				166MHz 1.8V
[F21] PC_15	eMMC_CMD	USDHC CMD	Input Output	<NA>				166MHz 1.8V
[G22] PD_00	eMMC_DATA0	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[E20] PD_01	eMMC_DATA1	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H19] PD_02	eMMC_DATA2	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H20] PD_03	eMMC_DATA3	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H18] PD_04	eMMC_DATA4	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D18] PD_05	eMMC_DATA5	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F19] PD_06	eMMC_DATA6	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F20] PD_07	eMMC_DATA7	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D20] PD_08	eMMC_RST	USDHC RST	Output	<NA>				166MHz 1.8V
[G19] PD_09	N.C	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[D19] PD_10	eMMC_DQS	USDHC DQS	Input	<NA>	No			166MHz 1.8V

Not applicable [✔ Approved]

4.27.1.1.2 Production

SM-72885 - MCU GPIOs init state Production

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[W10] PA_02	BOOTMOD0	BOOT	Input	<NA>	PD			50MHz 3.3V
[W11] PA_03	BOOTMOD1	BOOT	Input	<NA>	PD			50MHz 3.3V
[Y11] PA_07	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[U10] PA_08	EQ_IOs.EQ0_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[B8] PA_09	EQ_IOs.EQ1_3V3_GPIO_A5_PD_ACK	GPIO	Input	<NA>	PD			50MHz 3.3V
[E13] PA_10	DEBUG.3v3_DETECT	GPIO	Input	<NA>				50MHz 3.3V
[D8] PA_11	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C7] PA_12	CAN.3v3_B2B_EN	GPIO	Output	Low				50MHz 3.3V
[U12] PA_13	N.U	GPIO	High-Z	<NA>	PD			1MHz 3.3V
[W13] PA_15	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[W12] PB_00	MCU_IF.3v3_I2C0_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[E7] PB_01	MCU_IF.3v3_I2C0_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[D7] PB_02	WAKE.3v3_MCU_KEEPALIVE	GPIO	Output	Low				50MHz 3.3V
[AA10] PB_09	CAN.3v3_B2B_TX	CAN	Output	<NA>			500Kbps	50MHz 3.3V
[E6] PB_14	CAN.3v3_B2B_RX	CAN	Input	<NA>			500Kbps	50MHz 3.3V
[A6] PB_05	MCU_IF.3v3_I2C2_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[G9] PB_06	MCU_IF.3v3_I2C2_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[F7] PB_08	SAFETY_SIG.EQ0_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[U18] PH_02	SAFETY_SIG.3v3_Main_BAT_Fault_INT	EIRQ	Input	<NA>	PD			83MHz 3.3V
[V11] PB_10	SAFETY_SIG.3v3_Red_BAT_Fault_INT	EIRQ	Input	<NA>	PD			50MHz 3.3V
[G8] PB_11	MCU_IF.FR_TXEN	GPIO	Output	Low				50MHz 3.3V
[F8] PB_12	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[G7] PB_13	N.U	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[Y9] PA_06	EFUSE_EN	GPIO	Output	Low				50MHz 3.3V
[B5] PB_15	SAFETY_SIG.EQ1_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[V9] PC_00	DISCHARGE.ETH_MCU	GPIO	Output	Low				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[Y8] PC_01	MCU_IF.3v3_I2C4_SDA	I2C SDA	Input-Output	<NA>		OD	400kHz	50MHz 3.3V
[W8] PC_02	MCU_IF.3v3_I2C4_SCL	I2C SCL	Output	<NA>		OD	400kHz	50MHz 3.3V
[U9] PC_03	SAFETY_SIG.EQ1_3v3_NCRERR	EIRQ	Input	<NA>				50MHz 3.3V
[C13] PC_04	TEMP_SENS_ALERT	EIRQ	Input	<NA>				50MHz 3.3V
[G10] PC_05	ENC.3v3_UART_RX_R	GPIO	Input	<NA>			115Kbps	50MHz 3.3V
[A14] PC_06	ENC.3v3_UART_TX_R	GPIO	Input	<NA>				50MHz 3.3V
[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[B6] PC_08	PMIC_INTB	EIRQ	Input	<NA>				50MHz 3.3V
[U11] PC_09	MCU_IF.3v3_MCU_CLI_UART_TX	GPIO	High-Z	<NA>	PD		115Kbps	50MHz 3.3V
[Y12] PC_10	MCU_IF.3v3_MCU_CLI_UART_RX	GPIO	High-Z	<NA>	External PU		115Kbps	50MHz 3.3V
[F15] PC_11	CAN.3v3_HCP5_RX	CAN	Input	<NA>			2Mbps	50MHz 3.3V
[G11] PC_12	CAN.3v3_HCP5_TX	CAN	Output	<NA>			2Mbps	50MHz 3.3V
[Y7] PC_13	CAN.3v3_HCP5_EN	GPIO	Input Output	Low				50MHz 3.3V
[Y10] PD_11	EQ_IOs.EQ0_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[V23] PD_12	EQ_IOs.EQ0_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[R19] PD_13	EQ_IOs.EQ1_3v3_POR	GPIO	Output	Low				150MHz 3.3V
[N23] PH_00	EQ_IOs.EQ0_1_1v8_GPIO_B14_SYNC	GPIO	Output	Low				150MHz 1.8V
[P23] PD_14	ENC.1v8_PORZ	GPIO	Output	Low				150MHz 1.8V
[P20] PE_02	ETH.ENC_3v3_RGMII1_RX_CLK	GPIO	Output	Low				100MHz 3.3V
[U23] PE_03	ETH.ENC_3v3_RGMII1_RX_CTL	GPIO	Output	Low				150MHz 3.3V
[T23] PE_04	ETH.ENC_3v3_RGMII1_RD0	GPIO	Output	Low				150MHz 3.3V
[T22] PE_05	ETH.ENC_3v3_RGMII1_RD1	GPIO	Output	Low				150MHz 3.3V
[U22] PE_06	ETH.ENC_3v3_RGMII1_RD2	GPIO	Output	Low				150MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[T21] PE_07	ETH.ENC_3v3_RGMII1_RD3	GPIO	Output	Low				150MHz 3.3V
[R21] PE_08	ETH.ENC_3v3_RGMII1_TX_CLK	RGMII CLK	Input	<NA>				150MHz 3.3V
[R23] PE_09	ETH.ENC_3v3_RGMII1_TX_CTL	RGMII CTL	Input	<NA>				150MHz 3.3V
[P19] PE_10	ETH.ENC_3v3_RGMII1_TX_D0	RGMII DATA	Input	<NA>				150MHz 3.3V
[P18] PE_11	ETH.ENC_3v3_RGMII1_TX_D1	RGMII DATA	Input	<NA>				150MHz 3.3V
[N18] PE_12	ETH.ENC_3v3_RGMII1_TX_D2	RGMII DATA	Input	<NA>				150MHz 3.3V
[U21] PE_13	ETH.ENC_3v3_RGMII1_TX_D3	RGMII DATA	Input	<NA>				150MHz 3.3V
[T18] PE_14	CAN.3v3_RADAR1_STB	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[L22] PF_01	MCU_IF.1v8_MDIO	MDIO	Input Output	<NA>				150MHz 1.8V
[V17] PE_15	CAN.3v3_RADAR5_STB	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[M19] PF_00	MCU_IF.1v8_MDC	MDC	Output	<NA>				150MHz 1.8V
[Y21] PF_02	CAN.3v3_RADAR5_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[AA12] PA_14	SAFETY_SIG.DC_EQs_IN	GPIO	Output	High				50MHz 3.3V
[V14] PF_03	TDA4_EFUSE_EN	GPIO	Output	Low				150MHz 1.8V
[V13] PF_04	ENC.1v8_SoC_BOOTMODE0	GPIO	Output	Low				150MHz 1.8V
[U8] PF_15	MCU_CTL.power_EN_n	GPIO	Output	Low				50MHz 3.3V
[T19] PH_01	SAFETY_SIG.SW1_3v3_HARTBEAT	EIRQ	Input	<NA>				150MHz 3.3V
[L21] PD_15	ENC.1v8_MCU_PORZ	EIRQ	Output	Low				150MHz 1.8V
[U20] PH_03	VBAT_INPUT_STAGE.3V3_DIODE_PG	EIRQ	Input	<NA>	PU			150MHz 3V3
[P21] PE_00	EQ_IOs.EQ1_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[V21] PH_04	CAN.3v3_RADAR3_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W23] PH_05	VBAT_INPUT_STAGE.3V3_MAIN_HSS_STATUS	EIRQ	Input	<NA>	PU			150MHz 3.3V
[T20] PH_06	CAN.3v3_RADAR4_EN	GPIO	High-Z	<NA>	PD			50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[Y23] PH_07	CAN.3v3_RADAR4_STB	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[W21] PH_08	CAN.3v3_RADAR2_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[M23]PE_01	EQ_IOs.EQ0_1v8_GPIO_B21_XMODEM	GPIO	Output	Low				150MHz 1.8V
[W22] PH_09	CAN.3v3_RADAR2_STBn	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[V20] PH_10	CAN.3v3_RADAR1_EN	GPIO	High-Z	<NA>	PD			83MHz 3.3V
[U19] PJ_00	CAN.3v3_RADAR3_EN	GPIO	High-Z	<NA>	PD			50MHZ 3.3V
[B11] PJ_01	CAN.3v3_RADAR2_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D17] PJ_02	CAN.3v3_RADAR2_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D10] PJ_03	CAN.3v3_RADAR3_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C15] PJ_04	CAN.3v3_RADAR3_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C11] PJ_05	CAN.3v3_RADAR4_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[D15] PJ_06	CAN.3v3_RADAR4_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B15] PK_06	SAFETY_SIG.PMIC_3v3_SS	GPIO	Input	<NA>				50MHz 3.3V
[D12] PJ_07	CAN.3v3_RADAR1_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[D11]PK_05	VBAT_INPUT_STAGE.3v3_DIODE_RED_PG	EIRQ	Input	<NA>	PU			50MHz 3.3V
[F16] PJ_08	CAN.3v3_RADAR3_ERR	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[B12] PJ_09	MCU_CTL.SW0_RST	GPIO	High-Z	High				50MHz 3.3V
[E10] PK_09	EQ_IOs.3v3_MCU_EQ0_TX	CAN	Output	<NA>				50MHz 3.3V
[E14] PK_10	EQ_IOs.3v3_MCU_EQ0_RX	CAN	Input	<NA>				50MHz 3.3V
[E12] PJ_11	CAN.3v3_DEBUG_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[E16] PJ_12	CAN.3v3_DEBUG_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[A11] PJ_13	CAN.3v3_RADAR5_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[C16] PJ_14	CAN.3v3_RADAR5_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[A7] PL_04	CAN.3v3_B2B_STBn	GPIO	Output	Low				50MHz 3.3V
[F11] PK_01	EQ_IOs.3v3_MCU_EQ1_TX	CAN	Output	<NA>				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[D14] PK_02	EQ_IOs.3v3_MCU_EQ1_RX	CAN	Input	<NA>				50MHz 3.3V
[C6] PB_03	MCU_IF.3v3_I2C1_SCL	I2C SCL	Output	<NA>		OD	400KHz	50MHz 3.3V
[F14] PK_04	IGN_SENSE	GPIO	Input	<NA>				50MHz 3.3V
[E8] PB_04	MCU_IF.3v3_I2C1_SDA	I2C SDA	Input Output	<NA>		OD		50MHz 3.3V
[A15] PK_00	CAN.3v3_HCP5_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[C10] PK_07	EQ_IOs.EQ1_3v3_GPIO_A6_PD_REQ	GPIO	Output	Low				50MHz 3.3V
[F13] PK_08	SAFETY_SIG.VMs_3v3_INT	EIRQ	Input	<NA>				50MHz 3.3V
[A10] PJ_15	CAN.3v3_HCP5_STBn	GPIO	Input Output	Low				50MHz 3.3V
[E11] PK_03	HCP5_INH	GPIO	Input	<NA>				50MHz 3.3V
[A9] PK_11	DEBUG.3v3_MCU_UART_TX	GPIO	High-Z	<NA>	PD		115Kbps	50MHz 3.3V
[C14] PK_12	DEBUG.3v3_MCU_UART_RX	GPIO	High-Z	<NA>	External PU		115Kbps	50MHz 3.3V
[C9] PK_13	CAN.3v3_RADAR1_TX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[B13] PK_14	CAN.3v3_RADAR1_RX	GPIO	High-Z	<NA>	PD		500Kbps	50MHz 3.3V
[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	UART	Output	<NA>			921Kbps	50MHz 3.3V
[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[E9] PL_01	SAFETY_SIG.3v3_ENC_PMIC_SS	EIRQ	Input	<NA>				50MHz 3.3V
[A8] PL_02	MCU_CTL.3v3_ENC_PMIC_RST_OUT	EIRQ	Input	<NA>				50MHz 3.3V
[A13] PL_03	SAFETY_SIG.EQ0_3v3_CRERR	EIRQ	Input	<NA>				50MHz 3.3V
[D16] PJ_10	CAN.3v3_RADAR2_ERRn	GPIO	High-Z	<NA>	PD			50MHz 3.3V
[C8] PL_05	CAN.3v3_B2B_ERRn	GPIO	Input	<NA>	PU			50MHz 3.3V
[D13] PL_06	B2B_CAN_INH	GPIO	Input	<NA>				50MHz 3.3V
[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	UART	Input	<NA>			921Kbps	50MHz 3.3V
[N19] PL_08	CAN.1v8_RADAR4_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[A5] PB_07	SAFETY_SIG.DC_EQ1_OUT	GPIO	Input	<NA>				50MHz 3.3V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[L23] PL_09	CAN.1v8_RADAR5_ERR	GPIO	High-Z	<NA>	PD			50MHz 1.8V
[P22] PL_10	ENC.1v8_MCU_TIMESYNC0	GPIO	Input	<NA>	PD			166MHz 1.8V
[N22] PL_11	ENC.1v8_MCU_TIMESYNC1	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[N20] PL_12	SAFETY_SIG.SW0_1V8_EFR	EIRQ	Input	<NA>				150MHz 1.8V
[N21] PL_13	SAFETY_SIG.SW1_1V8_EFR	EIRQ	Input	<NA>				150MHz 1.8V
[M21] PL_14	SAFETY_SIG.SW0_1V8_HARTBEAT	EIRQ	Input	<NA>				166MHz 1.8V
[K20] PG_00	QSPI_A_SCK	QSPI CK	Output	<NA>			100MHz	133MHz 1.8V
[K21] PG_01	N.C	QSPI_A_CK	High-Z	<NA>	PD			150MHz 1.8V
[J21] PG_02	N.C	QSPI_A_CK2	High-Z	<NA>	PD			150MHz 1.8V
[H21] PG_03	N.C	QSPI_A_CK2	High-Z	<NA>	PD			150MHz 1.8V
[G21] PG_04	QSPI_A_CS0	QSPI_A_CS0	Output	Low				133MHz 1.8V
[H22] PG_05	TEST_POINT	QSPI_A_CS1	Output	Low				150MHz 1.8V
[L18] PF_05	QSPI_A_DATA0	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L19] PF_06	QSPI_A_DATA1	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[L20] PF_07	QSPI_A_DATA2	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K22] PF_08	QSPI_A_DATA3	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K19] PF_09	QSPI_A_DATA4	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[J23] PF_10	QSPI_A_DATA5	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[H23] PF_11	QSPI_A_DATA6	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K18] PF_12	QSPI_A_DATA7	QSPI DATA	Input Output	<NA>	PD			133MHz 1.8V
[K23] PF_13	QSPI_A_DQS	QSPI DQS	Input	<NA>	PD			133MHz 1.8V
[J19] PF_14	QSPI_A_INT	QSPI INT	Input	<NA>				133MHz 1.8V

Pin/Port#	Net Name	Mode	Init State	Init Value	Pull enable	OD/ PP	Data rates	Slew rate
[E19] PC_14	eMMC_CK	USDHC CK	Output	<NA>				166MHz 1.8V
[F21] PC_15	eMMC_CMD	USDHC CMD	Input Output	<NA>				166MHz 1.8V
[G22] PD_00	eMMC_DATA0	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[E20] PD_01	eMMC_DATA1	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H19] PD_02	eMMC_DATA2	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H20] PD_03	eMMC_DATA3	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[H18] PD_04	eMMC_DATA4	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D18] PD_05	eMMC_DATA5	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F19] PD_06	eMMC_DATA6	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[F20] PD_07	eMMC_DATA7	USDHC DATA	Input Output	<NA>				166MHz 1.8V
[D20] PD_08	eMMC_RST	USDHC RST	Output	<NA>				166MHz 1.8V
[G19] PD_09	N.C	GPIO	High-Z	<NA>	PD			166MHz 1.8V
[D19] PD_10	eMMC_DQS	USDHC DQS	Input	<NA>	No			166MHz 1.8V

Not applicable [✔ Approved]

4.27.2 GPIOs Expanders Init State

At power on, all ports are configured as input.

SM-80523 - Before setting the pins to output, the pin value required value to be set to Init Value. [✔ Approved]

4.27.2.1 GPIO expander

SM-72887 - DR64 GPIO expander

IC: U103

I2C: MCU_IF.3v3_I2C0

Address: 0x76

Pin/Port#	Net Name	Init State	Init Value	Notes
P00	ENC.3v3_MCU_BOOTMODE3	Input		
P01	ENC.3v3_MCU_BOOTMODE4	Input		

Pin/Port#	Net Name	Init State	Init Value	Notes
P02	ENC.3v3_MCU_BOOTMODE5	Input		
P03	ENC.3v3_SoC_BOOTMODE4	Input		
P04	ENC.3v3_SoC_BOOTMODE5	Input		
P05	MCU_CTL.3v3_ENC_PMIC_EN	Output	Low	
P06	ENC.3v3_RESET_REQZ	Output	Low	
P07	ENC.3v3_MCU_RESETZ	Output	Low	
P10	EN_RESET	Output	High	
P11	EQ_IOs.EQ1_3v3_GPIO_A7_MCU_GPIO0	Output	Low	
P12	EQ_IOs.EQ0_3v3_GPIO_A7_MCU_GPIO0	Output	Low	
P13	RADAR2_CAN_INH	Output	Low	
P14	ENC.3v3_SoC_BOOTMODE6	Input		
P15	EQ_IOs.EQ0_3v3_GPIO_A8_UBOOT	Output	Low	
P16	EQ_IOs.EQ1_3v3_GPIO_A8_UBOOT	Output	Low	
P17	FR_INH	Output	Low	

Not applicable [✔ Approved]

SM-72889 - DR64 GPIO expander
IC: U104
I2C: MCU_IF.3v3_I2C0
Address: 0x75

Pin/Port#	Net Name	Init State	Init Value
P00	DEBUG.10G_PHY_RST	Output	Low
P01	EQ_IOs.EQ0_3v3_FCMU_RST	Output	Low
P02	EQs_IOs.EQ0_3v3_GPIO_A14-TIMER0_EXT_ICAP1	Output	Low
P03	EQs_IOs.EQ1_3v3_GPIO_A14-TIMER0_EXT_ICAP1	Output	Low
P04	HEATER.3v3_LSS_EN	Output	Low
P05	MCU_CTL.I2B_5V_EN	Output	Low
P06	MCU_CTL.3v3_VDIs_CLKs_EN	Input	
P07	MCU_CTL.3v3_EQ0_EQ1_CG_DC_CLK_EN	Input	
P10	HEATER.3v3_LSS_STATUS	Output	Low
P11	MCU_EEPROM_WP	Input	
P12	EQ_IOs.EQ1_3v3_FCMU_RST	Output	Low
P13	USS.3v3_LSS_EN	Output	Low
P14	USS.3v3_LSS_STATUS	Output	Low
P15	HEATER.3v3_HSS_EN	Output	Low

Pin/Port#	Net Name	Init State	Init Value
P16	HEATER.3v3_HSS_DIAG_EN	Output	Low
P17	B2B_GPIO	Input	

Not applicable [✔ Approved]

SM-72890 - DR64 GPIO expander
IC: U146
I2C: MCU_IF.3v3_I2C0
Address: 0x74

Pin/Port#	Net Name	Init State	Init Value
P00	EQ_IOs.EQ0_3V3_GPIO_A12_MCU_GPIO1	Output	Low
P01	EQ_IOs.EQ0_3V3_GPIO_A15_MCU_INCAP	Output	Low
P02	EQ_IOs.EQ0_3V3_GPIO_A16_MCU_OUTCMP	Output	Low
P03	EQ_IOs.EQ0_3V3_GPIO_A17_MCU_GPIO2	Output	Low
P04	EQ_IOs.EQ1_3V3_GPIO_A12_MCU_GPIO1	Output	Low
P05	EQ_IOs.EQ1_3V3_GPIO_A15_MCU_INCAP	Output	Low
P06	EQ_IOs.EQ0_3v3_XINT	Input	
P07	EQ_IOs.EQ1_3v3_XINT	Input	
P10	EQ_IOs.EQ1_3V3_GPIO_A16_MCU_OUTCMP	Output	Low
P11	EQ_IOs.EQ1_3V3_GPIO_A17_MCU_GPIO2	Output	Low
P12	MCU_CTL.3v3_USS_CLK_EN	Input	
P13	MCU_CTL.3v3_EQ1_CLKs_EN	Input	
P14	MCU_CTL.3v3_EQ0_CLKs_EN	Input	
P15	MCU_CTL.12v_PoC_EN	Output	Low
P16	MCU_CTL.SW1_RST	Output	High
P17	VBAT_INPUT_STAGE.3v3_MAIN_HSS_EN	Output	Low

[✔ Approved]

Control Register

Following the successful acknowledgment of the address byte, the bus controller sends a command byte shown in

0	0	0	0	0	B2	B1	B0

Figure 8-6. Control Register Bits

Table 8-3. Command Byte

CONTROL REGISTER BITS			COMMAND BYTE (HEX)	REGISTER	PROTOCOL	POWER-UP DEFAULT
B2	B1	B0				
0	0	0	0x00	Input Port 0	Read byte	xxxx xxxx
0	0	1	0x01	Input Port 1	Read byte	xxxx xxxx
0	1	0	0x02	Output Port 0	Read-write byte	1111 1111
0	1	1	0x03	Output Port 1	Read-write byte	1111 1111
1	0	0	0x04	Polarity Inversion Port 0	Read-write byte	0000 0000
1	0	1	0x05	Polarity Inversion Port 1	Read-write byte	0000 0000
1	1	0	0x06	Configuration Port 0	Read-write byte	1111 1111
1	1	1	0x07	Configuration Port 1	Read-write byte	1111 1111

Configuration registers

The Configuration registers (registers 6 and 7) shown below configure the directions of the I/O pins. If a bit in this register is set to 1, the corresponding port pin is enabled as an input with a high-impedance output driver. If a bit in this register is cleared to 0, the corresponding port pin is enabled as an output.

Table 8-7. Registers 6 And 7 (Configuration Registers)

Bit	**C0.7**	**C0.6**	**C0.5**	**C0.4**	**C0.3**	**C0.2**	**C0.1**	**C0.0**
Default	1	1	1	1	1	1	1	1
Bit	**C1.7**	**C1.6**	**C1.5**	**C1.4**	**C1.3**	**C1.2**	**C1.1**	**C1.0**
Default	1	1	1	1	1	1	1	1

Output Port registers

The Output Port registers (registers 2 and 3) shown below show the outgoing logic levels of the pins defined as outputs by the Configuration register. Bit values in this register have no effect on pins defined as inputs. In turn, reads from this register reflect the value that is in the flip-flop controlling the output selection, not the actual pin value.

Table 8-5. Registers 2 And 3 (Output Port Registers)

Bit	**O0.7**	**O0.6**	**O0.5**	**O0.4**	**O0.3**	**O0.2**	**O0.1**	**O0.0**
Default	1	1	1	1	1	1	1	1
Bit	**O1.7**	**O1.6**	**O1.5**	**O1.4**	**O1.3**	**O1.2**	**O1.1**	**O1.0**
Default	1	1	1	1	1	1	1	1

SM-72891 -

Setting the Pins as output: Following the successful acknowledgment of the address byte, a command byte of 0x06(PORT0)/0x07(PORT1) is send. By this you can configure the PORT0/PORT1. Then set the bits of register 6 or 7 as 0 to configure as output and 1 to configure it as 0. **[** ✔ Approved **]**

4.28 DDR Configuration Report

4.28.1 General Information

DDR Type : *LPDDR4*
Firmware version : *2020_06_sp2*

4.28.2 UI Configuration

SM-80525 - Device Information :

- **Clock Cycle Freq (MHz)** : *1600 MHz*
- **DDR_CLK frequency** : *800MHz*
- **Density per channel (Gb)** : *16G*
- **Number of ROW Addresses** : *17*
- **Number of Chip Selects used** : *1*
- **Number of Channels** : *2*
- **Number of COLUMN Addresses** : *10*
- **Number of BANK addresses** : *3*
- **Number of BANKS** : *8*
- **Total DRAM density (Gb)** : *32*
- **Bus Width** : *32*
- **Clock Cycle Time (ns)** : *0.625*

[✔ Approved **]**

SM-80526 - Initial Board Settings :

- **PHY ODT Impedance** : *40 Ohm*

- **PHY Drive Strength** : *40 Ohm*
- **PHY Vref Quotient** : *0x13*
- **PHY Vref** : *0.163 V*
- **DRAM ODT Impedance** : *40 Ohm*
- **DRAM Drive Strength** : *40 Ohm*
- **CA Vref Training** : *yes*

[✔ Approved **]**

SM-80527 - ECC options :

- **Enable Inline ECC** : *no*
- **Note** : *Enabling ECC provides SEC/DED functionality - single bit error correction and double bit error detection.*
- **Train 2D** : *yes*
- **PHY Log Level** : *Firmware complete*
- **Refresh Command Constraints** : *Legacy*
- **Static Refresh Rate [0.25x]** : *no*
- **Per Bank Refresh** : *no*

[✔ Approved **]**

SM-80528 - DQ Swapping :

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
18	16	23	22	17	21	19	20	27	30	28	29	31	26	24	25	14	12	11	8	10	9	15	13	7	2	5	4	0	6	3	1

[✔ Approved **]**

SM-80529 - Code Generation :

Code format : *ATF* **[**✔ Approved **]**

SM-80530 - Advanced Settings :

Disable PMIC watchdog : *no* **[**✔ Approved **]**

4.28.3 Mode Registers

SM-80531 - Mode Register

<table>
<tr><th>#</th><th>value</th><th>OP [7]</th><th>OP [6]</th><th>OP [5]</th><th>OP [4]</th><th>OP [3]</th><th>OP [2]</th><th>OP [1]</th><th>OP [0]</th></tr>
<tr><td>MR1</td><td>0x54</td><td>RPST = 0x0</td><td colspan="3">nWR (for AP) = 0x5</td><td>RD-PRE = 0x0</td><td>WR-PRE = 0x1</td><td colspan="2">BL = 0x0</td></tr>
<tr><td>MR2</td><td>0x2D</td><td>WR Lev = 0x0</td><td>WLS = 0x0</td><td colspan="3">WL = 0x5</td><td colspan="3">RL = 0x5</td></tr>
<tr><td>MR3</td><td>0x33</td><td>DBI-WR = 0x0</td><td>DBI-RD = 0x0</td><td colspan="3">PDDS = 0x6</td><td>PPRP = 0x0</td><td>WR PST = 0x1</td><td>PU-CAL = 0x1</td></tr>
<tr><td>MR4</td><td>0x00</td><td>TUF = 0x0</td><td colspan="2">Thermal Offset = 0x0</td><td>PPRE = 0x0</td><td>SR Abort = 0x0</td><td colspan="3">Refresh Rate = 0x0</td></tr>
<tr><td>MR11</td><td>0x66</td><td>Reserved = 0x0</td><td colspan="3">CA ODT = 0x6</td><td>Reserved = 0x0</td><td colspan="3">DQ ODT = 0x6</td></tr>
<tr><td>MR12</td><td>0x4D</td><td>CBT Mode for Byte Mode = 0x0</td><td>VR-CA = 0x1</td><td colspan="6">Vref(CA) = 0xd</td></tr>
</table>

#	value	OP [7]	OP [6]	OP [5]	OP [4]	OP [3]	OP [2]	OP [1]	OP [0]
MR13	0x8	FSP-OP = 0x0	FSP-WR = 0x0	DMD = 0x0	RRO = 0x0	VRCG = 0x1	VRO = 0x0	RPT = 0x0	CBT = 0x0
MR14	0x4F	RFU = 0x0	VR(DQ) = 0x1	Vref(DQ) = 0xf					
MR16	0x00	PASR Bank Mask = 0x0							
MR17	0x00	PASR Segment Mask = 0x0							
MR22	0x6	ODTD for x8_2ch(Byte) mode = 0x0		ODTD-CA = 0x0	ODTE-CS = 0x0	ODTE-CK = 0x0	SOC ODT = 0x6		
MR24	0x00	TRR Mode = 0x0	TRR Mode BAn = 0x0			Unlimited = 0x0	MAC Value = 0x0		

[✔ Approved]

5 EyeQ6H

SM-80532 - The boot parameters of main and surround EyeQ is given below

SuppliersManagement/EyeQ6H Boot Parameters_Primary/EyeQ6H Boot Parameters_Primary_Main_B2

SuppliersManagement/EyeQ6H Boot Parameters_Primary/EyeQ6H Boot Parameters_Primary_Surround_B2 **[** In Review **]**

6 System Power Up Sequence

Comprehensive Power up and Down Sequence of a Board with MCU program steps:

If there is no number in Step column, then should ignore the row

6.1 Main_1/Degrade Mode

SM-63346 - Step 0

- Configure all IO Expander pins as per 4.27.2 GPIOs Expanders Init State
- Set IO Expander Pins to output: IO_ENC.3v3_MCU_BOOTMODE3, IO_EQ_IOs.EQ0_3v3_FCMU_RST, IO_EQ_IOs.EQ1_3v3_FCMU_RST, IO_ENC.3v3_RESET_REQZ, IO_ENC.3v3_MCU_RESETZ
- Set GPIO Pins to output: PortConf_PortPin_ETH.ENC_3v3_RGMII1_RX_CLK, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RX_CTL, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD0, PortConf_PortPin_ETH_ENC_3v3_RGMII1_RD1, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD2, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD3
- Set GPIO Pins mode to ALT2: PortConf_PortPin_ETH.ENC_3v3_RGMII1_RX_CLK, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RX_CTL, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD0, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD1, PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD2,
- Set GPIO Pins mode to ALT3: PortConf_PortPin_ETH.ENC_3v3_RGMII1_RD3

[✔ Approved]

SM-63347 - Main_1/Degraded Mode sequence

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timout
0	1	Assert WAKE.3v3_MCU_KEEPALIVE output	High			
0	2	Configure temperature sensor U60(0X48) using MCU_IF.3v3_I2C0				
0	3	Configure temperature sensor U62(0X49) using MCU_IF.3v3_I2C0				
0	4	Configure temperature sensor U64(0X4A) using MCU_IF.3v3_I2C0				
0	5	Configure temperature sensor U61(0X4C) using MCU_IF.3v3_I2C0				
0	6	Read DEBUG.3v3_DETECT to check the UART debug connected	High	Execute Step 8		
0	7	Read DEBUG.3v3_DETECT to check the UART debug connected	Low	Skip Step 8		
0	8	Assert DEBUG.10G_PHY_RST output	High	NA for Production		
0	9	Assert MCU_CTL.SW0_RST output	High			
0	10	Assert MCU_CTL.SW1_RST output	High			
0	11	Read SAFETY_SIG.SW0_1v8_EFR	High			1ms max
0	12	Read SAFETY_SIG.SW1_1v8_EFR	High			1ms max

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timout
0	13	Read HCP5_INH	Low	Oring of these signals		
0	14	Read IGN_SENSE	Low			
0	15	Read B2B_CAN_INH	Low			
0	16	Read VBAT_INPUT_STAGE.3V3_DIODE_PG	High	Oring of these signals		20ms max
0	17	Read VBAT_INPUT_STAGE.3V3_DIODE_RED_PG	High			20ms max
0	18	Read VBAT_INPUT_STAGE.MAIN_IN_HSS_VMON	Check ADC table	Oring of these signals		
0	19	Read VBAT_INPUT_STAGE.RED_IN_HSS_VMON	Check ADC table		60s	
0	20	Read VBAT_INPUT_STAGE.MCU_HSS_IMON	Check ADC table			
0	21	Read SAFETY_SIG.MAIN_GND_IMON	Check ADC table	Oring of these signals		
0	22	Read SAFETY_SIG.RED_GND_IMON	Check ADC table			
0	23	Read VR5510_AMUX	Check ADC table			
0	24	Enable VR5510(U54) spread spectrum MOD_EN -> set to 1 I2C bus #4 : MCU_IF.3v3_I2C4 PMIC main address : 0x20 Register M_CLOCK1 : Address 0x08 I2C write value : 0x10	MOD_EN bit set to HIGH			
0	25	Enable MPF53(U50) spread spectrum FSS_EN -> set to 1 I2C bus #0 : MCU_IF.3v3_I2C0 address : 0x28 Register CLK_CTRL : Address 0x11 I2C write value : 0x1b	FSS_EN bit set to HIGH			
0	26	Read 5v_AO_LDO_M	Check VM#3 table			
0	27	Read 0v8_SW0	0.8V ±5%			
0	28	Read 0v8_SW1	0.8V ±5%			
0	29	Read ETH_1v8	1.8V ±5%			
0	30	Read 1v2_SW0	1.2V ±5%			
0	31	Read 1v2_SW1	1.2V ±5%			
0	32	If DEBUG.3v3_DETECT =	High	Execute 32 and 33		
0	32	If DEBUG.3v3_DETECT =	Low	Skip 32 and 33		
0	33	Read 0V84_10G_PHY	0.84V ±5%	NA for Production		
0	34	Read 1V2_10G_PHY	1.2V ±5%	NA for Production		
0	35	Assert MCU_CTL.power_EN_n output	High		10ms	

Sequ ence	Step	Action	Status	Operations/ Dependency	Delay	Timout
0	36	Assert EN_RESET output	High			

[✓ Approved]

SM-63345 - MCU_CTL.SW0_RST and MCU_CTL.SW1_RST should set to High value before setting the pin mode to push-pull and output. [✓ Approved]

6.2 Main_2 Sequence

SM-63360 - Main_2 sequence

Seq uen ce	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	0	Assert SAFETY_SIG.DC_EQs_IN output	Low	After completion of seq 1		1ms
1	1	Read SAFETY_SIG.DC_EQ1_OUT	Low			
1	2	Assert SAFETY_SIG.DC_EQs_IN output	High			1ms
1	3	Read SAFETY_SIG.DC_EQ1_OUT	High			
1	4	Assert VBAT_INPUT_STAGE.3v3_MAIN_HSS_EN output	High			85ms
1	5	Read VBAT_INPUT_STAGE.MAIN_HSS_VMON	Check ADC table			85ms max
1	6	Read VBAT_INPUT_STAGE.MAIN_HSS_IMON	Check ADC table			80ms max
1	7	Read VBAT_INPUT_STAGE.3v3_MAIN_HSS_STATUS	High			25ms min to 100ms max
1	8	Assert MCU_CTL.power_EN_n output	Low			50ms
1	9	Read 5V	5V±5%			20ms max
1	10	Read 3v3	3.3V±5%			2ms max
1	11	Read 1V8 using U93	1.8V±5%			2ms max
1	12	Read DES_1V2	1.2V±5%			10ms max
1	13	Read DES_0V8	0.8V ±5%			2ms max
1	14	Read EQ0_VDD_CORE	0.875V+2.5% / -3%			2ms max
1	15	Read EQ0_VDDQ	0.5V±5%			5ms max
1	16	Read EQ0_1V05	1.05V±3%			5ms max
1	17	Read EQ0_1v8_IP	1.8V±3%			16ms max
1	18	Read EQ0_1V8_PCIe	1.8V±5%			17ms max
1	19	Read EQ0_0V75	0.75V±5%			5ms max
1	20	Read EQ0_0V86	0.86V±5%			18ms max
1	21	Read EQ1_VDD_CORE	0.875V+2.5% / -3%			2ms max
1	22	Read EQ1_VDDQ	0.5V±5%			5ms max

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	23	Read EQ1_1V05	1.05V±3%			5ms max
1	24	Read EQ1_1v8_IP	1.8V±3%			16ms max
1	25	Read EQ1_1V8_PCIe	1.8V±5%			17ms max
1	26	Read EQ1_0V75	0.75V±5%			5ms max
1	27	Read EQ1_0V86	0.86V±5%			18ms max
1	28	Read TDA4_0v8_CPU_AVS	0.8V±5%			10ms
1	29	Read TDA4_0v8_VDD_CORE	0.8V±5%			10ms
1	30	Read TDA_1V1	1.1V±5%			18ms max
1	31	Read TDA4_0v85	0.85V±5%			7ms
1	32	Read TDA4_1v8_PHY_LDO	1.8V±5%			7ms
1	33	Read TDA4_0v8_DLL_LDO	0.8V±5%			7ms
1	34	Read TDA4_1v8_PLL_LDO	1.8v±5%			7ms
1	35	Read 12vPoC_M	Check VM#3 table			10ms max
1	36	Assert MCU_CTL.I2B_5V_EN output	High			
1	37	Read 5V_AUD1	5V±5%			5ms
1	38	Enable 3V3(U33) spread spectrum SSCEN -> set to 1 I2C bus #1 : MCU_IF.3v3_I2C1 Device address : 0x33 Register CONTROL1 : Address 0x01 I2C write value : 0x6a	SSCEN set to HIGH			
1	39	Enable 1V8(U32) spread spectrum SSCEN -> set to 1 I2C bus #2 : MCU_IF.3v3_I2C2 Device address : 0x30 Register CONTROL1 : Address 0x01 I2C write value : 0x6a	SSCEN set to HIGH			
1	40	Enable EQ0_VDD_CORE(A_U100)spread spectrum. SSCEN -> set to 1 I2C bus #1 : MCU_IF.3v3_I2C1 Device address : 0x46 Register CONTROL1 : Address 0x01 I2C write value : 0x68	SSCEN set to HIGH			
1	41	Enable EQ1_VDD_CORE(B_U100)spread spectrum. SSCEN -> set to 1 I2C bus #2 : MCU_IF.3v3_I2C2 Device address : 0x46 Register CONTROL1 : Address 0x01 I2C write value : 0x68	SSCEN set to HIGH			

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	42	Enable EQ0_1V05 (A_U18)spread spectrum. SSCEN -> set to 1 I2C bus #1 : MCU_IF.3v3_I2C1 Device address : 0x43 Register CONTROL1 : Address 0x01 I2C write value : 0x6a	SSCEN set to HIGH			
1	43	Enable EQ0_VDDQ (A_U26)spread spectrum. SSCEN -> set to 1 I2C bus #1 : MCU_IF.3v3_I2C1 Device address : 0x40 Register CONTROL1 : Address 0x01 I2C write value : 0x6a	SSCEN set to HIGH			
1	44	Enable EQ1_VDDQ (B_U26)spread spectrum. SSCEN -> set to 1 I2C bus #2 : MCU_IF.3v3_I2C2 Device address : 0x40 Register CONTROL1 : Address 0x01 I2C write value : 0x6a	SSCEN set to HIGH			
1	45	Enable DES_0V8(U34)spread spectrum. SSCEN -> set to 1 I2C bus #2 : MCU_IF.3v3_I2C2 Device address : 0x43 Register CONTROL1 : Address 0x01 I2C write value : 0x6a	SSCEN set to HIGH			
1		Configure VMs				
1	46	Configure VM #1 (U93) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x37 Register BANK_SEL : Address 0xF0 I2C write value : 0x01 (Bank 1) Register IEN_CONTROL : Address 0x1B I2C write value : 0x15 Register IEN_TEST : Address 0x1C I2C write value : 0x09				
1	47	Configure VM #2 (U94) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x36 Register BANK_SEL : Address 0xF0 I2C write value : 0x01 (Bank 1) Register IEN_CONTROL : Address 0x1B I2C write value : 0x15 Register IEN_TEST : Address 0x1C I2C write value : 0x09				

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	48	Configure VM #3 (U95) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x35 Register BANK_SEL : Address 0xF0 I2C write value : 0x01 (Bank 1) Register IEN_CONTROL : Address 0x1B I2C write value : 0x15 Register IEN_TEST : Address 0x1C I2C write value : 0x09				
1	49	Configure VM #4 (U114) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x34 Register BANK_SEL : Address 0xF0 I2C write value : 0x01 (Bank 1) Register IEN_CONTROL : Address 0x1B I2C write value : 0x15 Register IEN_TEST : Address 0x1C I2C write value : 0x09				
1	50	Configure VM #5 (U148) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x33 Register BANK_SEL : Address 0xF0 I2C write value : 0x01 (Bank 1) Register IEN_CONTROL : Address 0x1B I2C write value : 0x15 Register IEN_TEST : Address 0x1C I2C write value : 0x09				
		Clear all VM				

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	51	Configure VM #1 (U93) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x37 Register BANK_SEL : Address 0xF0 I2C write value : 0x00 (Bank 0) Register INT_CONTROL : Address 0x1B I2C write value : 0xff Register INT_TEST : Address 0x1C I2C write value : 0xff Register INT_UVHF: Address 0x12 I2C write value : 0xff Register INT_UVLF: Address 0x14 I2C write value : 0xff Register INT_OVHF: Address 0x16 I2C write value : 0xff Register INT_OVLF: Address 0x18 I2C write value : 0xff				
1	52	Configure VM #2 (U94) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x36 Register BANK_SEL : Address 0xF0 I2C write value : 0x00 (Bank 0) Register INT_CONTROL : Address 0x1B I2C write value : 0xff Register INT_TEST : Address 0x1C I2C write value : 0xff Register INT_UVHF: Address 0x12 I2C write value : 0xff Register INT_UVLF: Address 0x14 I2C write value : 0xff Register INT_OVHF: Address 0x16 I2C write value : 0xff Register INT_OVLF: Address 0x18 I2C write value : 0xff				

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	53	Configure VM #3 (U95) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x35 Register BANK_SEL : Address 0xF0 I2C write value : 0x00 (Bank 0) Register INT_CONTROL : Address 0x1B I2C write value : 0xff Register INT_TEST : Address 0x1C I2C write value : 0xff Register INT_UVHF: Address 0x12 I2C write value : 0xff Register INT_UVLF: Address 0x14 I2C write value : 0xff Register INT_OVHF: Address 0x16 I2C write value : 0xff Register INT_OVLF: Address 0x18 I2C write value : 0xff				
1	54	Configure VM #4 (U114) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x34 Register BANK_SEL : Address 0xF0 I2C write value : 0x00 (Bank 0) Register INT_CONTROL : Address 0x1B I2C write value : 0xff Register INT_TEST : Address 0x1C I2C write value : 0xff Register INT_UVHF: Address 0x12 I2C write value : 0xff Register INT_UVLF: Address 0x14 I2C write value : 0xff Register INT_OVHF: Address 0x16 I2C write value : 0xff Register INT_OVLF: Address 0x18 I2C write value : 0xff				

Sequence	Step	Action	Status	Operations/ Dependency	Delay	Timeout
1	55	Configure VM #5 (U148) I2C bus #0 : MCU_IF.3v3_I2C0 Device address : 0x33 Register BANK_SEL : Address 0xF0 I2C write value : 0x00 (Bank 0) Register INT_CONTROL : Address 0x1B I2C write value : 0xff Register INT_TEST : Address 0x1C I2C write value : 0xff Register INT_UVHF: Address 0x12 I2C write value : 0xff Register INT_UVLF: Address 0x14 I2C write value : 0xff Register INT_OVHF: Address 0x16 I2C write value : 0xff Register INT_OVLF: Address 0x18 I2C write value : 0xff				
1	56	Clear fault interrupt of 0x37 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2				
1	57	Clear fault interrupt of 0x36 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2				
1	58	Clear fault interrupt of 0x34 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2				
1	59	Clear fault interrupt of 0x35 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2				
1	60	Clear fault interrupt of 0x33 address of MCU_IF.3v3_I2C0 using INT_SCR1 and INT_SCR2				
1	61	Read SAFETY_SIG.VMs_3v3_INT	High			

[✔ Approved]

6.3 Final

SM-63361 - Final

Sequence	Step DR64	Action	Status	Operations/Dependency	Delay	Timeout
		Power, clocks, bootstraps				
2	0	Read MCU_CTL.3v3_ENC_PMIC_RST_OUT	High	After completion of seq 2		100ms max
2	1	Assert MCU_CTL.3v3_EQ0_CLKs_EN output	Low			
2	2	Assert MCU_CTL.3v3_EQ1_CLKs_EN output	Low			

Sequence	Step DR64	Action	Status	Operations/Dependency	Delay	Time out
2	3	Assert MCU_CTL.3v3_VDIs_CLKs_EN output	Low			
2	4	Assert MCU_CTL.3v3_EQ0_EQ1_CG_DC_CLK_EN output	Low			
2	5	Assert EQ_IOs.EQ0_3v3_XINT output	High			
2	6	Assert EQ_IOs.EQ1_3v3_XINT output	High			
2	7	Assert EQ_IOs.EQ0_1v8_GPIO_B21_XMODEM output	Low			
2	8	Assert EQ_IOs.EQ1_1v8_GPIO_B21_XMODEM output	Low			
2	9	Assert EQ_IOs.EQ0_3v3_GPIO_A8_UBOOT output	Low			
2	10	Assert EQ_IOs.EQ1_3v3_GPIO_A8_UBOOT output	Low			
2	11	Assert ENC.3v3_MCU_BOOTMODE3 output	High			
2	12	Assert ENC.3v3_MCU_BOOTMODE4 output	High			
2	13	Assert ENC.3v3_MCU_BOOTMODE5 output	Low			
2	14	Assert ENC.1v8_SoC_BOOTMODE0 output	Low			
2	15	Assert ENC.3v3_SoC_BOOTMODE4 output	High			
2	16	Assert ENC.3v3_SoC_BOOTMODE5 output	Low			
2	17	Assert ENC.3v3_SoC_BOOTMODE6 output	High			
		Daisy chain check after EQ PWR enabled				
2	18	Assert SAFETY_SIG.DC_EQs_IN output	Low			1ms
2	19	Read SAFETY_SIG.DC_EQ1_OUT	Low			
2	20	Assert SAFETY_SIG.DC_EQs_IN output	High			1ms
2	21	Read SAFETY_SIG.DC_EQ1_OUT	High			
		Ready for deasserting reset signals				
2	22	Assert EN_RESET output	Low		1ms	
2	23	Assert EQ_IOs.EQ0_3v3_POR output	High		1ms	
2	24	Assert EQ_IOs.EQ0_3v3_FCMU_RST output	High			
2	25	Assert EQ_IOs.EQ1_3v3_POR output	High		1ms	
2	26	Assert EQ_IOs.EQ1_3v3_FCMU_RST output	High			
2	27	Assert ENC.1v8_PORZ output	High			
2	28	Assert ENC.1v8_MCU_PORZ output	High		1ms	
2	29	Assert ENC.3v3_RESET_REQZ output	High			
2	30	Assert ENC.3v3_MCU_RESETZ output	High		1ms	

Sequence	Step DR64	Action	Status	Operations/Dependency	Delay	Time out
2	31	Set below pins as input ENC.1v8_SoC_BOOTMODE0 ENC.3v3_SoC_BOOTMODE4 ENC.3v3_SoC_BOOTMODE5 ENC.3v3_SoC_BOOTMODE6 ENC.3v3_MCU_BOOTMODE3 ENC.3v3_MCU_BOOTMODE4 ENC.3v3_MCU_BOOTMODE5			20ms	
2	32	Read TEMP_SENS_ALERT	High			
2	33	If DEBUG.3v3_DETECT =	High	Step 36 to 41		
2	34	If DEBUG.3v3_DETECT =	Low	Skip step 36 to 41		
2	35	Assert ENABLE_UART_EQ_MAIN using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p00	High	NA for Production		
2	36	Assert ENABLE_UART_EQ_SUROUND using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p01	High	NA for Production		
2	37	Assert ENABLE_UART_ENCODER using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p02	High	NA for Production		
2	38	Assert ENABLE_UART_EQ_MAIN using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p00	Low	NA for Production		
2	39	Assert ENABLE_UART_EQ_SUROUND using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p01	Low	NA for Production		
2	40	Assert ENABLE_UART_ENCODER using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p02	Low	NA for Production		
2	41	configure pin ENC.3v3_UART_TX_R to UART				
2	42	configure pin ENC.3v3_UART_RX_R to UART				
2	43	Read SAFETY_SIG.PMIC_3v3_SS	High			
		EyeQ6_ 0&1 BIST monitoring				
2	44	Wait for the BIST occurrence				250ms max
2	45	Read SAFETY_SIG.EQ0_3v3_NCRERR	Toggle			
2	46	Read SAFETY_SIG.EQ1_3v3_NCRERR	Toggle			
2	47	Read SAFETY_SIG.EQ0_3v3_CRERR	Toggle			
2	48	Read SAFETY_SIG.EQ1_3v3_CRERR	Toggle			

[✔ Approved]

7 System Power down Sequence

7.1 Main Power down sequence

SM-64686 -

Step	IC	RefDes	MCU port	Input Net	Output Net	Status	Action	Delay	Dependency
0							assert high ENABLE_UART_EQ_MAIN using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p00	NA for Production	
0							assert high ENABLE_UART_EQ_SUROUND using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p01	NA for Production	
0							assert high ENABLE_UART_ENCODER using U1 MCU_IF.3v3_I2C0_SCL /SDA 0x77 p02	NA for Production	

[Approved]

SM-64688 - Main

Step	Action	Status	Condition to go next step	wait time to go next step	Timeout
1	Assert MCU_CTL.I2B_5V_EN	Low			
2	Assert EQ_IOs.EQ1_3v3_GPIO_A6_PD_REQ	High			10ms
3	Assert EQ_IOs.EQ0_3v3_GPIO_A6_PD_REQ	High			10ms
	wait for EYEQ6_1 graceful shutdown complete				< 10s
	wait for EYEQ6_0 graceful shutdown complete				
4	Wait for EQ_IOs.EQ1_3V3_GPIO_A5_PD_ACK		ACK == high		
5	Wait for EQ_IOs.EQ0_3V3_GPIO_A5_PD_ACK		ACK == high		
6	Assert EQ_IOs.EQ1_3v3_FCMU_RST	Low			
7	Assert EQ_IOs.EQ0_3v3_FCMU_RST	Low			
8	Assert ENC.3v3_RESET_REQZ	Low			
			EQs are OFF, TDA4 after Shutdown Request		
9	Assert EN_RESET output	Low			
	Assert EQ0_POR_n	Low	Internal HW step		
	Assert EQ_IOs.EQ1_3v3_POR	Low	Internal HW step		
	Assert ENC.1v8_PORZ	Low	Internal HW step		
	Assert ENC.1v8_MCU_PORZ	Low	Internal HW step		

Step	Action	Status	Condition to go next step	wait time to go next step	Timeout
	Assert EQ_IOs.EQ0_3v3_GPIO_A4_TIMER1_CLK_PDB_0	Low	Internal HW step		
	Assert EQ_IOs.EQ1_3v3_GPIO_A4_TIMER1_CLK_PDB_0	Low	Internal HW step		
10	Assert MCU_CTL.3v3_EQ1_CLKs_EN	High	disable = high impedance /high		10ms
11	Assert MCU_CTL.3v3_EQ0_CLKs_EN	High	disable = high impedance /high		10ms
12	Assert MCU_CTL.3v3_EQ0_EQ1_CG_DC_CLK_EN	High	disable = high impedance /high		10ms
13	Assert MCU_CTL.3v3_VDIs_CLKs_EN	High	disable = high impedance /high		10ms
14	Assert MCU_CTL.power_EN_n	High		5ms	
15	Read EQ0_1v8_IP		read value <=0.8V		
16	Read EQ0_1V8_PCIe		read value <=0.8V		
17	Read EQ0_0V86		read value <=0.3V		
18	Read EQ0_0V75		read value <=0.3V		
19	Read EQ1_1v8_IP		read value <=0.8V		
20	Read EQ1_1V8_PCIe		read value <=0.8V		
21	Read EQ1_0V86		read value <=0.3V		
22	Read EQ1_0V75		read value <=0.3V		
23	Read EQ0_VDD_CORE		read value <=0.2V		
24	Read EQ0_1V05		read value <=0.3V		
25	Read EQ0_VDDQ		read value <=0.3V		
26	Read EQ1_VDD_CORE		read value <=0.2V		
27	Read EQ1_1V05		read value <=0.3V		
28	Read EQ1_VDDQ		read value <=0.3V		
29	Read TDA4_0v8_VDD_CORE		read value <=0.5V		
30	Read TDA4_0v8_CPU_AVS		read value <=0.5V		
31	Read TDA4_1v1		read value <=0.5V		
32	Read TDA4_0v85		read value <=0.5V		
33	Read TDA4_1v8_PHY_LDO		read value <=0.8V		
34	Read TDA4_0v8_DLL_LDO		read value <=0.5V		
35	Read TDA4_1v8_PLL_LDO		read value <=0.8V		
36	Read DES_1V2		read value <=0.5V		
37	Read 12vPoC_M (not applicable)		read value <=1V		
38	Read 5V		read value <=1V		
39	Read 3v3		read value <=0.3V		
40	Read 1v8 using U93		read value <=1V		5ms max
41	Assert VBAT_INPUT_STAGE.3v3_MAIN_HSS_EN	Low			30ms

Step	Action	Status	Condition to go next step	wait time to go next step	Timeout
42	Assert MCU_CTL.SW1_RST	Low			
43	Assert MCU_CTL.SW0_RST	Low			
44	Assert DEBUG.10G_PHY_RST	Low	NA for production		
45	Assert DISCHARGE.ETH_MCU	High		0ms	
46	Assert -VR5510 PMIC bit **PWRON1DIS** Register M_MODE - Address 0x01 PMIC main address : 0x20 I2C bus #4	High	disable wake signal		
47	Assert -VR5510 PMIC bit **GOTO_OFF** Register M_SM_CTRL1 - Address 0x02 PMIC main address : 0x20 I2C bus #4	High	shutdown PMIC		

[In Review]

7.2 Degraded mode power down sequence

SM-64690 - Degraded Mode

Step	Action	Status	Condition to go next step	wait time to go next step
0	Assert MCU_CTL.SW1_RST	Low		
1	Assert MCU_CTL.SW0_RST	Low		
2	Assert DEBUG.10G_PHY_RST	Low	NA for production	
3	Assert DISCHARGE.ETH_MCU	High		0ms
4	Assert HIGH -VR5510 PMIC bit **PWRON1DIS** Register M_MODE - Address 0x01 PMIC main address : 0x20 I2C bus #4		disable wake signal	
5	Assert HIGH -VR5510 PMIC bit **GOTO_OFF** Register M_SM_CTRL1 - Address 0x02 PMIC main address : 0x20 I2C bus #4		shutdown PMIC	

[Approved]

8 Reset requirement

8.1 Resetting of TDA4

SM-64697 - While performing the TDA4 reset below step are done

Step	MCU port	Output Net	Status	Action	Required delay before next step	
0				Set below pin as input ENC.3v3_UART_RX_R Set below pin as input ENC.3v3_UART_TX_R		
1	I²C0 U103 P00	ENC.3v3_MCU_BOOTMODE3	High	Encoder MCU Bootmode 3 set		
2	I²C0 U103 P01	ENC.3v3_MCU_BOOTMODE4	High	Encoder MCU Bootmode 4 set		
3	I²C0 U103 P02	ENC.3v3_MCU_BOOTMODE5	Low	Encoder MCU Bootmode 5 set		
4	I²C0 U103 P03	ENC.3v3_SoC_BOOTMODE4	High	Encoder SoC Bootmode 4 set		
5	I²C0 U103 P04	ENC.3v3_SoC_BOOTMODE5	Low	Encoder SoC Bootmode 5 set		
6	PF04	ENC.1v8_SoC_BOOTMODE0	Low	Encoder SoC Bootmode 0 set		
7	I²C0 U103 P14	ENC.3v3_SoC_BOOTMODE6	High	Encoder SoC Bootmode 6 set		
8	PD14	ENC.1v8_PORZ	Low	Assert low ENC.1v8_PORZ		
9	PD15	ENC.1v8_MCU_PORZ	Low	Assert low ENC.1v8_MCU_PORZ	1ms	
10	I²C0 U103 P06	ENC.3v3_RESET_REQZ	Low	Assert low ENC.3v3_RESET_REQZ		
11	I²C0 U103 P07	ENC.3v3_MCU_RESETZ	Low	Assert low ENC.3v3_MCU_RESETZ	1ms	
12	PD14	ENC.1v8_PORZ	High	Assert high ENC.1v8_PORZ		
13	PD15	ENC.1v8_MCU_PORZ	High	Assert high ENC.1v8_MCU_PORZ	1ms	
14	I²C0 U103 P06	ENC.3v3_RESET_REQZ	High	Assert high ENC.3v3_RESET_REQZ		
15	I²C0 U103 P07	ENC.3v3_MCU_RESETZ	High	Assert high ENC.3v3_MCU_RESETZ	1ms	
16				Set below pins as input ENC.1v8_SoC_BOOTMODE0 ENC.3v3_SoC_BOOTMODE4 ENC.3v3_SoC_BOOTMODE5 ENC.3v3_MCU_BOOTMODE3 ENC.3v3_MCU_BOOTMODE4 ENC.3v3_MCU_BOOTMODE5 ENC.3v3_SoC_BOOTMODE6	20ms	

Step	MCU port	Output Net	Status	Action	Required delay before next step	
17				Set below pins as UART ENC.3v3_UART_RX_R ENC.3v3_UART_TX_R		

[✔ Approved]

8.2 Resetting of EyeQs

SM-64694 - While performing the EyeQ reset below step are done

Step	Output Net	Action	Status	Required delay before next step
0	EQ_IOs.EQ0_3v3_FCMU_RST	Assert EQ_IOs.EQ0_3v3_FCMU_RST	Low	
1	EQ_IOs.EQ0_3v3_POR	Assert EQ_IOs.EQ0_3v3_POR	Low	
2	EQ_IOs.EQ1_3v3_FCMU_RST	Assert EQ_IOs.EQ1_3v3_FCMU_RST	Low	1ms
3	EQ_IOs.EQ1_3v3_POR	Assert EQ_IOs.EQ1_3v3_POR	Low	6ms
4	EQ_IOs.EQ0_3v3_POR	Assert EQ_IOs.EQ0_3v3_POR	High	
5	EQ_IOs.EQ0_3v3_FCMU_RST	Assert EQ_IOs.EQ0_3v3_FCMU_RST	High	
6	EQ_IOs.EQ1_3v3_POR	Assert EQ_IOs.EQ1_3v3_POR	High	
7	EQ_IOs.EQ1_3v3_FCMU_RST	Assert EQ_IOs.EQ1_3v3_FCMU_RST	High	

[✔ Approved]

8.3 Resetting of Ethernet Switch

SM-64695 - While performing the Ethernet Switch reset below step are done

Step	Output Net	Action	Status	Required delay before next step
0	MCU_CTL.SW1_RST	Assert MCU_CTL.SW1_RST	Low	
1	MCU_CTL.SW0_RST	Assert MCU_CTL.SW0_RST	Low	1ms
2	DEBUG.10G_PHY_RST	Assert DEBUG.10G_PHY_RST	Low	NA for production
3	MCU_CTL.SW1_RST	Assert High MCU_CTL.SW1_RST	High	
4	MCU_CTL.SW0_RST	Assert High MCU_CTL.SW0_RST	High	
5	DEBUG.10G_PHY_RST	Assert DEBUG.10G_PHY_RST	High	NA for production

[✔ Approved]

9 Revision history

Ver.	Date	Updated by	Section	Essence of change	Approved by	
0.1	22-07-2026	Chaithra Shetty	4.18.6	U148 register 0x1B 0x1C value change		
0.2	10-08-2026	Chaithra Shetty	4.26 4.27.1	ENC.1v8_MCU_TIMESYNC1 made NA ENC.1v8_MCU_TIMESYNC1 - GPIO init defined as HIGH-Z ,NA ,PD as not in use.	Frank Bergen Moshe Shalev	